



TILERA[®]

***BASEBOARD
MANAGEMENT
CONTROLLER USER'S
GUIDE***

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CHAPTER 1 ABOUT THE BMC

1.1 Overview

This manual accompanies platforms on which Baseboard Management Controller (BMC) software and hardware are installed. These platforms include the following:

- TILEmpower-Gx36™
- TILEmpower-Gx72™
- TILEmpower-Gx36 FR
- TILEmpower-Gx72 FR
- TILEmpower-Gx72 MP
- TILExtreme-Gx™
- TILExtreme-Gx™ Duo

The BMC provides three connectivity interfaces, including

- I²C for power monitoring, temperature, LEDs, etc.
- USB connectivity to the TILE-Gx™ processor including console over USB

Note: The BMC uses the USB for TILE-Gx console as well.

- An internal FE switch to the TILE-Gx processor

Note: This interface is currently unused.

Refer to the platform user's guide for a BMC block diagram of your platform.

1.2 BMC Features

The BMC management interface provides:

- Connectivity for console operations for the CPU module in the platform
- Access to management statistics and monitor logs for device sensors installed in the platform
- Support for Power Control through the web management interface and the `ipmitool` Command Line Interface (CLI)
- A mechanism to upgrade the BMC firmware image

1.3 Determining IP Addresses

Throughout this guide we mention IP addresses, which differ depending on the platform on which you have a BMC system (both hardware and software) installed.

Note: You must have the `ipmitool` binary installed on your Linux x86 host system to use the `ipmitool` command as described below. Run the `rpm -i` or `yum` commands to install the RPM package on your host machine.

1.3.1 Determining the IP Address of the BMC

You can determine the IP address for the BMC using the BMC console or an `ipmiutil` command.

1.3.1.1 Finding the BMC IP Address from the Console

At power-on, the BMC briefly displays its DHCP IP on this console, so you must watch for it carefully.

1.3.1.2 Finding the BMC IP Address Using IPMI Tools

The BMC provides limited support for the Intelligent Platform Management Interface (IPMI) version 2 protocols. You can use IPMI standard tools to query IPMI devices over Remote Management Control Protocol (RMCP). Use either an `ipmiutil discover` command or `idisc-`
`cover` command (which are most often the same) to query all IPMI-compliant devices on your network.

Use the `idiscover` command from the `ipmiutil` package to determine the IP address for the BMC. Here is an example of the IP address for the TILEmpower-Gx36 platform:

```
$ idiscover -b 10.8.0.130 -e 10.8.0.180
idiscover ver 1.8
Discovering IPMI Devices:
.....
01| response from | 10.8.0.128          |
..
02| response from | 10.8.0.150          |
```

CHAPTER 2 PERFORMING ADMINISTRATIVE FUNCTIONS

2.1 Overview

This chapter describes various procedures used to perform administrative functions that include the following:

- Setting up and maintaining user accounts
- Connecting to CPU module consoles
- Changing passwords
- Displaying Baseboard Management Controller (BMC) software release information

Most procedures can be completed using Command Line Interface (CLI) or GUI-based methods, as listed in [Table 2-1](#).

Table 2-1. Common BMC IPMI Commands and GUI Equivalents

IPMI Function	Description	GUI Function
Section C.2.3 “Displaying Firmware Release Information Using the BMC INFO Command” on page 43	Gives basic information about the BMC.	Section 2.5 “Displaying BMC Release Information” on page 11
Section C.2.4 “Changing the Network Configuration Using the LAN Command” on page 44	Changes the network configuration for eth0 and eth1 (internal network if available).	Section 3.4 “Changing the Network Configuration” on page 20
Section C.2.5 “Viewing all the IPMI Sensors Using the SDR Command” on page 44	Views all IPMI Sensors.	Section 4.2 “Viewing Sensor Data Captured by the BMC” on page 24
Section C.2.6 “Viewing Detailed Information on Sensors and Changing Alarm Thresholds Using the SENSOR Command” on page 46	Views detailed information on IPMI Sensors and change sensor alarm thresholds.	Section 4.3 “Viewing Sensor Alarm Thresholds” on page 26
Section C.2.7 “Printing System Event Logs Using the SEL Command” on page 47	Prints the System Event Log.	Section 4.4 “Printing System Event Logs” on page 27
Section C.2.8 “Configuring Event Filters Using the PEF Command” on page 47	Configures event filters, alert policies, and LAN destinations.	Section 4.5 “Configuring Event Filters” on page 28
Section C.2.9 “Managing User Accounts Using the USER Command” on page 47	Manages user accounts.	Section 2.4.3 “Managing User Accounts” on page 8

2.2 User Accounts and Access Levels

2.2.1 Using the Web GUI

For instructions, see [“Changing the BMC Admin Password” on page 10](#).

2.2.2 Using IPMI Commands

Use IPMI commands to specify user accounts and access levels as follows:

```
-bash-4.1$ ipmitool user
User Commands: summary [<channel number>]
list          <channel number>]
set name      <user id> <username>
set password  <user id> [<password>]
disable       <user id>
enable        <user id>
priv          <user id> <privilege level> [<channel number>]
test          <user id> <16|20> [<password>]
```

2.3 Connecting to CPU Module Consoles

Use `ssh` to login to the consoles for the CPU module in the platform. [Table 2-1](#) lists the mapping between CPU module number and the BMC username for the console.

Note: By default, there are no passwords for BMC console user accounts.

Specify the address as the IP address for the BMC interface on the platform, prepended by the console number (1 through 4) for the CPU module to which you want to connect. For example, to connect to the console on CPU #1:

```
$ ssh con1@ip_address
```

For systems with multiple TILE-Gx chips per CPU module (e.g. the TILEExtreme-Gx Duo), append `a` or `b` to connect to the specific TILE-Gx node.

`con1a`, `con1b`, `con2a`, `con2b`, etc...

The TILEmpower-Gx has one CPU, TILExtreme-Gx has four CPUs, and the TILExtreme-Gx Duo has eight CPUs with the correspondence between CPU number, BMC console username, and values to use for the `tile-monitor --dev` command listed in [Table 2-2](#).

Table 2-2. CPU Number and BMC Username Correspondence for the TILEmpower-Gx, TILExtreme-Gx, and TILExtreme-Gx Duo

CPU Number	Platform	Node Number	ssh Username	tile-monitor --dev Value
CPU #1	TILEmpower-Gx	1	con1	<code>bmc_ip_addr#1</code>
	TILExtreme-Gx Duo	1a	con1a	<code>bmc_ip_addr#1a</code>
		1b	con1b	<code>bmc_ip_addr#1b</code>
CPU #2	TILExtreme-Gx	2	con2	<code>bmc_ip_addr#2</code>
	TILExtreme-Gx Duo	2a	con2a	<code>bmc_ip_addr#2a</code>
		2b	con2b	<code>bmc_ip_addr#2b</code>
CPU #3	TILExtreme-Gx	3	con3	<code>bmc_ip_addr#3</code>
	TILExtreme-Gx Duo	3a	con3a	<code>bmc_ip_addr#3a</code>
		3b	con3b	<code>bmc_ip_addr#3b</code>
CPU #4	TILExtreme-Gx	4	con4	<code>bmc_ip_addr#4</code>
	TILExtreme-Gx Duo	4a	con4a	<code>bmc_ip_addr#4a</code>
		4b	con4b	<code>bmc_ip_addr#4b</code>

Note: We recommend that you use `tile-console` instead of `ssh`. While they perform the same functions, you should use the same tool for all console connectivity.

The example below shows how the command is expressed when using `tile-console --dev`:

```
tile-console --dev 192.168.0.15#3a
```

2.4 Changing the Passwords

This section describes how to change both BMC Administrator and System Administrator passwords.

2.4.1 Changing the BMC Admin Password

You can change the passwords for the four CPU console usernames using the following procedure.

1. Login to the BMC via the browser. Use the pathname provided for you. We recommend that you use a current release of your browser.
2. At the logon screen ([Figure 2-1](#)), enter the assigned username and password, and click **Logon** to display the **Dashboard**.

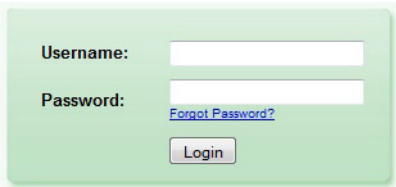
A green rectangular login form with a white border. It contains two input fields: 'Username:' and 'Password:'. Below the password field is a blue link labeled 'Forgot Password?'. At the bottom center is a 'Login' button.

Figure 2-1: Logon Screen

- Note:** To refresh the web display, should click the **Refresh** icon at the topmost-right menu bar (the third icon from right). Otherwise, clicking the browser’s refresh button forces a logout.
3. Choose the name of the user. From the Dashboard of the BMC, pictured in [Figure 2-2](#), choose **Configuration**→ **Users**. The **User Management** dialog appears.
 4. Choose a Username to Modify in the BMC. Choose admin from the **Username** row by clicking on the name.

User Management

The list below shows the current list of available users. To delete or modify a user, select their name in the list and press "Delete User" or "Modify User". To add a new user, select an unconfigured slot and press "Add User".

Number of configured users: 2

UserID ↕	Username ↕	User Access ↕	Network Privilege ↕	SNMP Status ↕	Email ID ↕
1	console	Enabled	Administrator	Disabled	~
2	admin	Enabled	Administrator	Disabled	~
3	~	~	~	~	~
4	~	~	~	~	~
5	~	~	~	~	~
6	~	~	~	~	~
7	~	~	~	~	~
8	~	~	~	~	~
9	~	~	~	~	~
10	~	~	~	~	~

Add UserModify UserDelete User

Figure 2-2: User Management Dialog

5. Modify user information. Click **Modify User**. The **Modify User** dialog appears.

Modify User

Username: admin

☐ Change Password

Password Size: ☒ 16 Bytes ☐ 20 Bytes

Password:

Confirm Password:

User Access: ☒ Enable

Network Privilege: Administrator

SNMP Status: ☐ Enable

SNMP Access: Read Only

Authentication Protocol: SHA

Privacy Protocol: DES

Email ID:

Email Format: AMI-Format

Uploaded SSH Key: Not Available

New SSH Key: No file selected.

Figure 2-3: Changing a Username Password in the BMC

- In the **Modify User** dialog, click the **Change Password** box.
- Choose a **Password Size**.
- Enter the new password into the **Password** field, and again into the **Confirm Password** field.
- Click **Modify**.

2.4.2 Changing the BMC System Administrator Password

You can change the passwords for the system administrator using the following procedure.

1. Login to the BMC using the browser. Use the pathname provided for you. We recommend that you use a current release of your browser.
2. At the logon screen ([Figure 2-1](#)), enter the assigned username and password (admin, admin), and click **Logon** to display the **Dashboard**.
3. Once the **Dashboard** is displayed, select **Maintenance** → **System Administrator** to display the **System Administrator** dialog, shown in [Figure 2-4](#).

System Administrator

This page allows the user to enable/disable access and change the password for the System Administrator account.

Username

sysadmin

User Access

☒ Enable

☐ Change Password

Password

Confirm Password

Save

Reset

Figure 2-4: System Administrator Dialog

4. Click on the **Change Password** option, specify the password in the **Password** field, enter the password again in the **Confirm Password** field, and click on **Save**.

2.4.3 Managing User Accounts

The **User Management** screen allows you to view the current list of user slots for the server. You can add a new user and modify or delete the existing users.

Log on to the BMC and select **Configuration**→**Users** (see [Figure 2-5](#)).

User Management

The list below shows the current list of available users. To delete or modify a user, select their name in the list and press "Delete User" or "Modify User". To add a new user, select an unconfigured slot and press "Add User".

Number of configured users: 2

UserID ↕	Username ↕	User Access ↕	Network Privilege ↕	SNMP Status ↕	Email ID ↕
1	console	Enabled	Administrator	Disabled	~
2	admin	Enabled	Administrator	Disabled	~
3	~	~	~	~	~
4	~	~	~	~	~
5	~	~	~	~	~
6	~	~	~	~	~
7	~	~	~	~	~
8	~	~	~	~	~
9	~	~	~	~	~
10	~	~	~	~	~

Add User

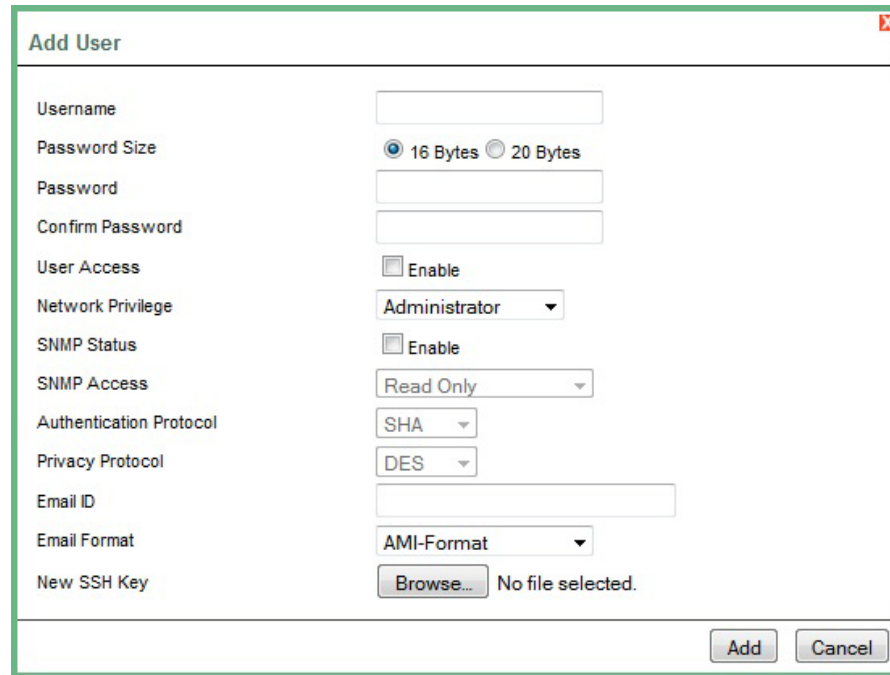
Modify User

Delete User

Figure 2-5: User Management Screen

To add a user:

1. Click on a UserID line where no username is listed, and click **Add User** to display the **Add User** dialog.



The **Add User** dialog box contains the following fields and controls:

- Username:** A text input field.
- Password Size:** Radio buttons for **16 Bytes** (selected) and **20 Bytes**.
- Password:** A text input field.
- Confirm Password:** A text input field.
- User Access:** A checkbox labeled **Enable**.
- Network Privilege:** A dropdown menu currently showing **Administrator**.
- SNMP Status:** A checkbox labeled **Enable**.
- SNMP Access:** A dropdown menu currently showing **Read Only**.
- Authentication Protocol:** A dropdown menu currently showing **SHA**.
- Privacy Protocol:** A dropdown menu currently showing **DES**.
- Email ID:** A text input field.
- Email Format:** A dropdown menu currently showing **AMI-Format**.
- New SSH Key:** A **Browse...** button and the text **No file selected.**

At the bottom right are **Add** and **Cancel** buttons.

Figure 2-6: Add User Dialog

2. Enter the name of the user in the **User Name** field.

The **User Name** is a string of 4 to 16 alpha-numeric characters that must start with an alphabetical character, be case-sensitive, and include special characters.

Note: The following characters are not allowed: ','(comma), '.'(period), ':'(colon), ';' (semicolon), ' '(space), '/'(slash), '\' (backslash), '[' (left bracket) and ']' (right bracket).

3. In the **Password** and **Confirm Password** fields, enter and confirm your new password.

Note: The Password must be at least 8 characters long, must not include white space and be no more than 20 characters in length.

4. In the **Network Privilege** field, enter the network privilege assigned to the user, indicated as: Administrator, Operator, User, or No Access.
5. In the **Email ID** field, enter the email ID of the user. If the user forgets the password, the new password is mailed to the configured email address.

Note: The SMTP Server must be configured to send emails.

6. In the **New SSH Key** field, click **Browse** and select the SSH key file.

To delete a user name:

Click on an entry line where a user is listed, and click **Delete User**.



Warning, deleting the `console` account results in CPU Console sessions being inoperable (see [Section 2.3 Connecting to CPU Module Consoles](#)).

To modify user privileges:

1. Click on an entry line where a user is listed, and click **Modify User** to display the **Modify User** dialog ([Figure 2-7](#)).

Figure 2-7: Modify User Dialog

2. Edit the required fields.
To change the password, enable the **Change Password** option.
3. After editing the changes, click **Modify** to return to the users list page or click **Cancel** at any time to exit the dialog.

2.5 Displaying BMC Release Information

The BMC version information is displayed on the **Dashboard**.

1. Login to the BMC via the browser. Use the pathname provided for you. We recommend that you use a current release of your browser.
2. At the logon screen (Figure 2-1), enter the assigned username and password, and click **Logon** to display the **Dashboard**.
3. Click on the **Dashboard** tab to display the **BMC Dashboard** (Figure 2-8).
4. View firmware release information under the **Device Information** heading.

TILERA®

admin (Administrator) Refresh

Dashboard FRU Information Server Health Configuration Remote Control Maintenance Firmware Update

Dashboard

Dashboard gives the overall information about the status of the device and remote server.

Device Information

Firmware Revision: 1.6.8
Firmware Build Time: Apr 21 2014 09:01:02 EDT

Network Information (Edit)

MAC Address: 00:1A:CA:00:C1:43
V4 Network Mode: DHCP
IPv4 Address: 10.4.0.241
V6 Network Mode: Disable

Remote Control Console not supported

Sensor Monitoring

Status	Sensor	Reading	
●	Chassis GPIO	0x8011	🔍
●	Chassis Avg	30 ° C	🔍
●	Aggregated Fault	0x803F	🔍
●	Fan Tray 1	6880 RPM	🔍
●	Fan Tray 2	6800 RPM	🔍
●	Fan Tray 3	6800 RPM	🔍
●	FanTray 1 Status	0x803E	🔍
●	FanTray 2 Status	0x803E	🔍
●	FanTray 3 Status	0x803E	🔍
●	FanRedundancy	0x8001	🔍

Figure 2-8: BMC Dashboard with Firmware Release Information

2.6 Determining IPMI Status on the Platform

You can determine the state of the platform using an IPMI chassis command as in the following example:

```
$ ipmitool -H <hostname or IP Address> -U admin -P admin chassis status
System Power           : on
Power Overload         : false
Power Interlock        : inactive
Main Power Fault       : false
Power Control Fault    : false
Power Restore Policy   : always-on
Last Power Event       : ac-failed
Chassis Intrusion      : inactive
Front-Panel Lockout    : inactive
Drive Fault            : false
Cooling/Fan Fault      : false
```

You can determine the state of the platform using an IPMI raw command as well, as with this example:

```
-bash-3.2$ ipmitool -H <hostname or IP Address> -U admin -P admin raw 0x38 0x00 02 ff
```

The BMC responds to the status request by sending back two hex bytes. The bit-masks for these bytes are the same as defined in [Section 5.4.1 Understanding the Bit Mask for an IPMI Raw Command](#) (but without a “0x” prefix). The first byte indicates which of the devices is currently powered up, with “1” corresponding to a device that is powered up. The second byte indicates which devices are in reset mode, where a “1” indicates that a device is operational (and thus not in reset).

The first byte of the response, 02, indicates that only CPU #1 is powered up; or in the case of TILExtreme-Duo, only CPUs #1a and #1b are powered up. The second byte of the response, ff, which indicates that none of the devices are in reset mode.

Note: For most uses, you want to get the response ff ff from an IPMI raw chassis status command — it indicates that every device is working.

2.7 Updating the BMC Firmware

To upload a new BMC firmware image into a Tiler platform, follow these directions.

Note: During this process we recommend that you open a BMC console window so you can watch the process as it is happening.

1. Save your SSL settings. From the BMC Dashboard, chose **Maintenance** → **Preserve Configuration**. The **Preserve Configuration** dialog appears ([Figure 2-9](#)).

Note: Tiler recommends that you save SSH (to maintain the same SSH key) and Authentication settings before you upgrade the BMC firmware image.

Preserve Configuration

This page allows the user to configure the preserve configuration items, which will be used by the Restore factory defaults to preserve the existing configuration without overwriting with default configuration.

Click here to go to [Firmware Update](#) or [Restore Factory Defaults](#)

Number of Preserved Items: 1

#	Preserve Configuration Item	Preserve Status
1	SDR	☐
2	FRU	☐
3	SEL	☐
4	IPMI	☐
5	Network	☐
6	NTP	☐
7	SSH	☒
8	KVM	☐
9	Authentication	☒

Check All

Uncheck All

Save

Reset

Preserve SSH box

Preserve Authentication box

Save

Figure 2-9: Preserving SSH Settings in the BMC ¹

2. Check the **Preserve Status** box in the line for SSH to preserve your SSH key settings.
3. Check the **Preserve Status** box in the line for Authentication to preserve your Authentication settings.
4. Click **Save**.
5. Either click **Firmware Update** on the **Preserve Configuration** dialog or choose **Firmware Update** → **Firmware Update**. The **Firmware Update** dialog appears.

Firmware Update

Upgrade firmware of the device. Press "Enter Update Mode" to put the device in update mode.

The protocol information to be used for firmware image transfer during this update is as follows. To configure, choose 'Protocol Configuration' under Firmware Update menu.

Protocol Type : HTTP/HTTPs

WARNING: Please note that after entering update mode widgets, other web pages and services will not work. All open widgets will be closed automatically. If upgrade process is cancelled in the middle of the wizard, the device will reset.

- ☒ Closing all active client requests.
- ☒ Preparing device for firmware upgrade.
- ☐ Uploading firmware image. ↻
- ☐ Verifying firmware image.
- ☐ Flashing firmware image.
- ☐ Resetting Device.

Upload Firmware

Please select the firmware image to flash

No file selected.

Upload SignImage Key

Enter Preserve Configuration

Enter Update Mode

Figure 2-10: Firmware Update through the BMC

6. Click **Enter Update Mode**. The **Firmware Update** dialog appears.
7. Click **OK** to enter the update mode.

You will not be able to perform any other tasks until firmware upgrade is complete and the device is rebooted. Click Ok if you want to enter the update mode?

1. Note: the KVM option (8) is not supported.

Note: If the unable to perform until firmware upgrade caution appears, click OK. For some reason the BMC is in a state that did not allow the firmware update to occur. This resets the BMC firmware. It is okay to try again.

Firmware Update

Upgrade firmware of the device. Press "Enter Update Mode" to put the device in update mode.

The protocol information to be used for firmware image transfer during this update is as follows. To configure, choose 'Protocol Configuration' under Firmware Update menu.

Protocol Type : HTTP/HTTPS

WARNING: Please note that after entering update mode widgets, other web pages and services will not work. All open widgets will be closed automatically. If upgrade process is cancelled in the middle of the wizard, the device will reset.

☒ Closing all active client requests.

☒ Preparing device for firmware upgrade.

☐ Uploading firmware image. ↻

☐ Verifying firmware image.

☐ Flashing firmware image.

☐ Resetting Device.

Upload Firmware

Please select the firmware image to flash

Browse...

No file selected.

Upload

Cancel

Upload SignImage Key

Enter Preserve Configuration

Enter Update Mode

Figure 2-11: Upload New Firmware File Dialog

	Warning: Do not interrupt the uploading process until it is completed. Once it is completed, the system automatically reboots, and you need to login to the server again.
---	---

As the software prepares for the process of upgrading software, the screen tracks the progress.

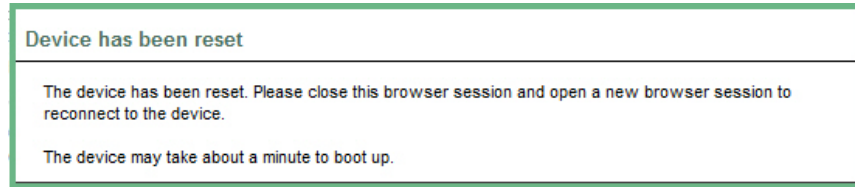
- 8. Click **Choose File**. Then navigate to, then select the *.ima file that you want to upload.
- 9. Click **Upload**. The **Firmware Version** dialog appears, showing the current and new image versions.
- 10. If the new firmware image version is correct, click **Proceed**.
- 11. Click **OK**. The BMC management interface provides you with status information on the progress of flashing the new firmware image into the BMC.

The BMC resets when you are finished uploading the firmware update.

Note: Only the BMC gets reset. The Gx nodes continue to run unaffected.

Note: if the **Firmware Update** dialog does not appear to be tracking the progress of flashing the image into the BMC, examine the RS-232 console connection. Until you see the BMC Linux console reset and reboot, do not cycle the power to the platform, or you will corrupt the BMC firmware. Some web browsers are reported to have issues in updating the screen during this process. The flashing process should take less than 10 minutes.

Once the update is complete, or if you cancel the firmware upgrade, the following message is displayed:



Do not power cycle the system until the BMC has fully initialized after resetting.

CHAPTER 3 CONFIGURING NETWORK OPTIONS

3.1 Overview

This chapter describes how BMC firmware enables users to configure SSL, to create passwords or disable the requirement to have password access to the BMC interface, to manage firewall configuration, and to create and manage user accounts and access levels.

Most procedures can be completed using Command Line Interface (CLI) or GUI-based methods, as listed in [Table 3-1](#).

Table 3-1. Common BMC IPMI Commands and GUI Equivalents

IPMI Function	Description	GUI Function
Section C.2.1 “Powering the Platform On/Off/Cycling or Resetting Using the CHASSIS POWER Command” on page 43	Powers the platform on/off/cycle or resets it.	Section 5.2.1 “Cycling Power on the Platform from the BMC” on page 30
Section C.2.2 “Causing the Chassis to Blink using the CHASSIS IDENTIFY Command” on page 43	Causes the chassis LEDs to blink for a specified period of time.	Not applicable
Section C.2.3 “Displaying Firmware Release Information Using the BMC INFO Command” on page 43	Gives basic information about the BMC.	Section 2.5 “Displaying BMC Release Information” on page 11
Section C.2.4 “Changing the Network Configuration Using the LAN Command” on page 44	Changes the network configuration for eth0 and eth1 (internal network if available).	Section 3.4 “Changing the Network Configuration” on page 20
Section C.2.5 “Viewing all the IPMI Sensors Using the SDR Command” on page 44	Views all IPMI Sensors.	Section 4.2 “Viewing Sensor Data Captured by the BMC” on page 24
Section C.2.6 “Viewing Detailed Information on Sensors and Changing Alarm Thresholds Using the SENSOR Command” on page 46	Views detailed information on IPMI Sensors and change sensor alarm thresholds.	Section 4.3 “Viewing Sensor Alarm Thresholds” on page 26
Section C.2.7 “Printing System Event Logs Using the SEL Command” on page 47	Prints the System Event Log.	Section 4.4 “Printing System Event Logs” on page 27
Section C.2.8 “Configuring Event Filters Using the PEF Command” on page 47	Configures event filters, alert policies, and LAN destinations.	Section 4.5 “Configuring Event Filters” on page 28
Section C.2.9 “Managing User Accounts Using the USER Command” on page 47	Manages user accounts.	Section 2.4.3 “Managing User Accounts” on page 8

3.2 Configuring SSL

Configure the Secure Socket Layer (SSL) layer to access the device in secured mode. This procedure involves uploading the SSL Certificate and private key file into the BMC in a secured manner.

Note: SSL Certificate can only be configured using the Web GUI.

1. Login to the BMC. Open your browser, and use the pathname provided for you to display the logon screen. We recommend that you use a current release of your browser.

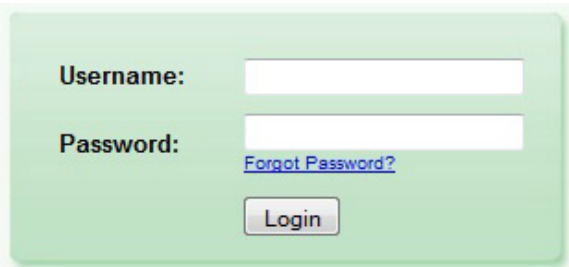
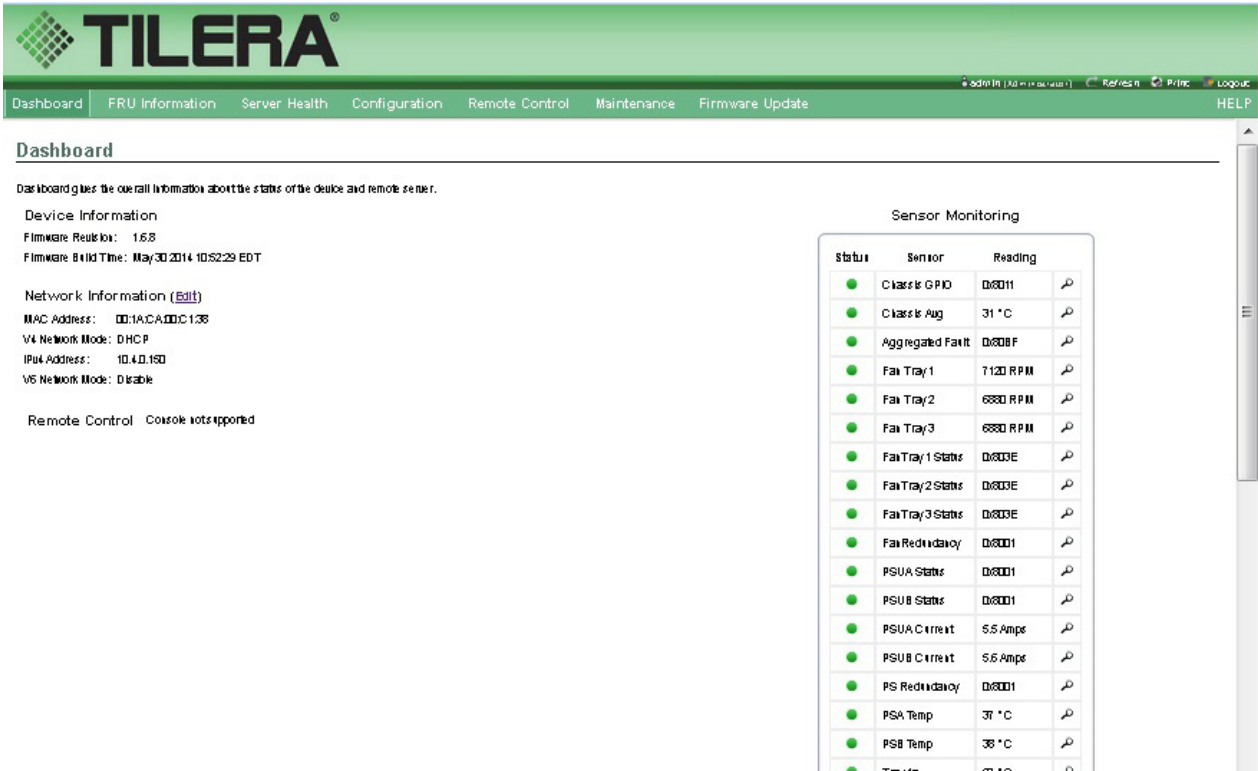


Figure 3-1: Logon Screen

2. At the logon screen (Figure 3-1), enter the assigned username and password (admin/admin), and click **Login** to display the **Dashboard** screen.



Status	Sensor	Reading	
●	Classik GPIO	0x0011	↻
●	Classik Avg	31 °C	↻
●	Aggregated Fan It	0x00FF	↻
●	Fan Tray1	1120 RPM	↻
●	Fan Tray2	6880 RPM	↻
●	Fan Tray3	6880 RPM	↻
●	Fan Tray1 Status	0x003E	↻
●	Fan Tray2 Status	0x003E	↻
●	Fan Tray3 Status	0x003E	↻
●	Fan Redundancy	0x0001	↻
●	PSUA Status	0x0001	↻
●	PSUB Status	0x0001	↻
●	PSUA Current	5.5 Amps	↻
●	PSUB Current	5.5 Amps	↻
●	PS Redundancy	0x0001	↻
●	PSA Temp	37 °C	↻
●	PSB Temp	38 °C	↻

Figure 3-2: Dashboard Screen

3. Access the **SSL Certificate Configuration** screen. From the **Dashboard** of the BMC, pictured in Figure 3-2, choose **Configuration** → **SSL**. The **SSL Certificate Configuration** dialog (Figure 3-3) appears.

Figure 3-3: SSL Certificate Configuration Screen

- Click on the **Upload SSL** tab if it is not already selected. Complete the fields as described below:

Current Certificate	Displays the date and time on which the certificate was uploaded.
New Certificate	Chooses a new certificate file. SSL certificate files are in PEM formats (not supported).
Current Privacy Key	Displays the date and time on which the Privacy Key is set.
New Privacy Key	Chooses a new Privacy Key file to be uploaded. Privacy Key files are in PEM format.

- Click the **Upload** key to upload the certificate and/or privacy key selected, or **Cancel** to cancel the operation.
- Upload the privacy key when a prompt appears to upload the privacy key.
If the key was uploaded successfully, a message indicating success appears.
- Click **OK** to reset BMC software.

You can now access your BMC software securely using the following format in your IP. For example, if your IP address is 192.168.0.30, enter the following:

`https://192.168.0.30`

Note: The <s> after <http> indicates that the web server is secured. You must accept the certificate before you can access the application.

3.3 Establishing a Firewall Using an AMI OEM Command

Once SSL is setup (port 443), an admin user can disable port 80 to require secure transactions to the Web GUI.

For more information, see “Supported OEM Commands” on page 51.

After you configure SSL to use port 443, you can disable port 80 to require that users connect to the Web GUI *only* using that secure method.

3.4 Changing the Network Configuration

1. Login to the BMC via the browser. Use the pathname provided for you. We recommend that you use a current release of your browser.
2. At the logon screen (Figure 3-1), enter the assigned username and password, and click **Logon** to display the **Dashboard**.
3. Click on the **Dashboard** tab to display the **BMC Dashboard** (Figure 3-4).

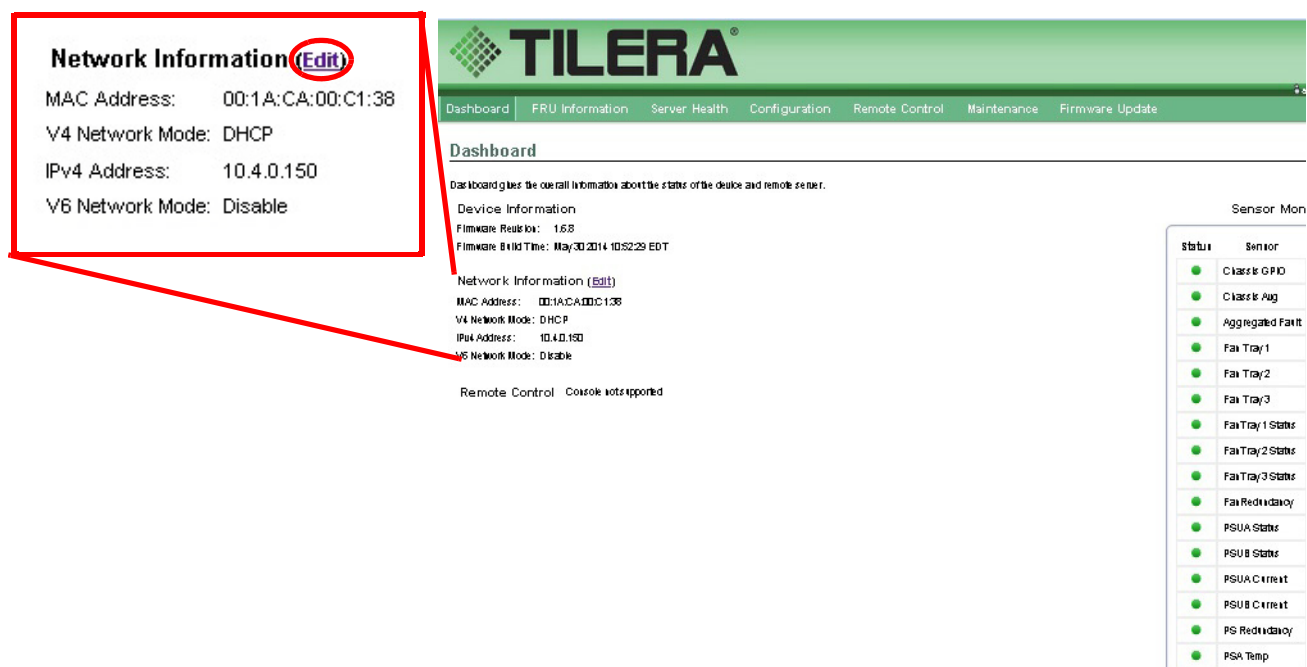
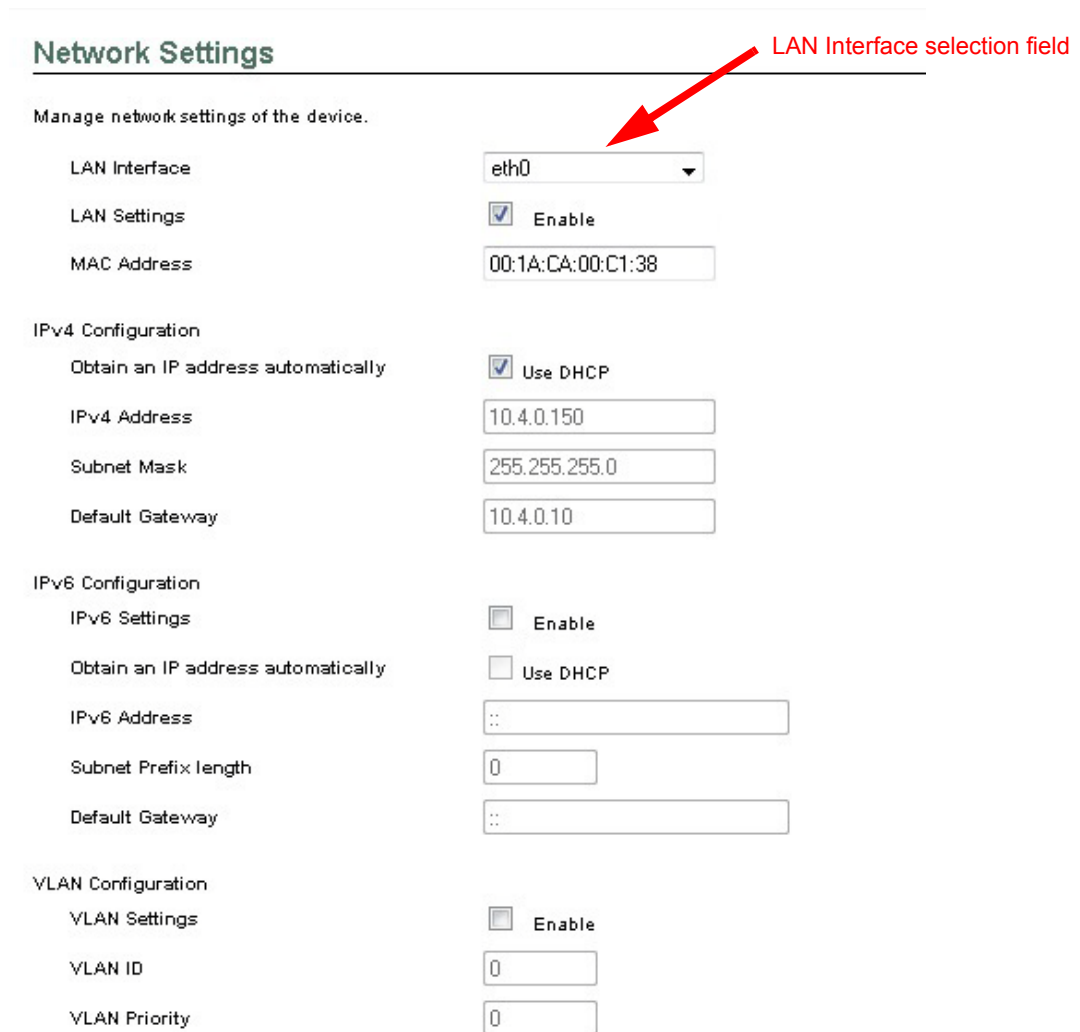


Figure 3-4: BMC Dashboard

4. Click on the **Edit** link (highlighted in Figure 3-4) to display the **Network Settings** screen.



Network Settings

Manage network settings of the device.

LAN Interface	eth0
LAN Settings	☒ Enable
MAC Address	00:1A:CA:00:C1:38

IPv4 Configuration

Obtain an IP address automatically	☒ Use DHCP
IPv4 Address	10.4.0.150
Subnet Mask	255.255.255.0
Default Gateway	10.4.0.10

IPv6 Configuration

IPv6 Settings	☐ Enable
Obtain an IP address automatically	☐ Use DHCP
IPv6 Address	::
Subnet Prefix length	0
Default Gateway	::

VLAN Configuration

VLAN Settings	☐ Enable
VLAN ID	0
VLAN Priority	0

A red arrow points from the text "LAN Interface selection field" to the "eth0" dropdown menu in the LAN Interface row.

Figure 3-5: Network Settings Screen

5. Click on the **LAN Interface** selection field to change the interface and then click **Save**.

CHAPTER 4 HANDLING SENSORS AND LOGS

4.1 Overview

This chapter describes how to review sensor data, including alarm thresholds and event logs.

Most procedures can be completed using Command Line Interface (CLI) or GUI-based methods, as listed in [Table 4-1](#).

Table 4-1. Common BMC IPMI Commands and GUI Equivalents

IPMI Function	Description	GUI Function
Section C.2.1 “Powering the Platform On/Off/Cycling or Resetting Using the CHASSIS POWER Command” on page 43	Powers the platform on/off/cycle or resets it.	Section 5.2.1 “Cycling Power on the Platform from the BMC” on page 30
Section C.2.2 “Causing the Chassis to Blink using the CHASSIS IDENTIFY Command” on page 43	Causes the chassis LEDs to blink for a specified period of time.	Not applicable
Section C.2.3 “Displaying Firmware Release Information Using the BMC INFO Command” on page 43	Gives basic information about the BMC.	Section 2.5 “Displaying BMC Release Information” on page 11
Section C.2.4 “Changing the Network Configuration Using the LAN Command” on page 44	Changes the network configuration for eth0 and eth1 (internal network if available).	Section 3.4 “Changing the Network Configuration” on page 20
Section C.2.5 “Viewing all the IPMI Sensors Using the SDR Command” on page 44	Views all IPMI Sensors.	Section 4.2 “Viewing Sensor Data Captured by the BMC” on page 24
Section C.2.6 “Viewing Detailed Information on Sensors and Changing Alarm Thresholds Using the SENSOR Command” on page 46	Views detailed information on IPMI Sensors and change sensor alarm thresholds.	Section 4.3 “Viewing Sensor Alarm Thresholds” on page 26
Section C.2.7 “Printing System Event Logs Using the SEL Command” on page 47	Prints the System Event Log.	Section 4.4 “Printing System Event Logs” on page 27
Section C.2.8 “Configuring Event Filters Using the PEF Command” on page 47	Configures event filters, alert policies, and LAN destinations.	Section 4.5 “Configuring Event Filters” on page 28
Section C.2.9 “Managing User Accounts Using the USER Command” on page 47	Manages user accounts.	Section 2.4.3 “Managing User Accounts” on page 8

4.2 Viewing Sensor Data Captured by the BMC

The BMC supports temperature sensor monitoring for all temperature sensors installed in the platform. The BMC presents all sensor data on the **Dashboard**.

This procedure describes how to view sensor data.

1. Click on the **Dashboard** tab to display the **BMC Dashboard**.
2. To view details for a specific sensor, click on the **Magnifying Glass** icon for that sensor listed in the table.

Note that what is displayed (specifically the number and kind of sensors) on the **Dashboard** differs depending on which platform you are accessing.

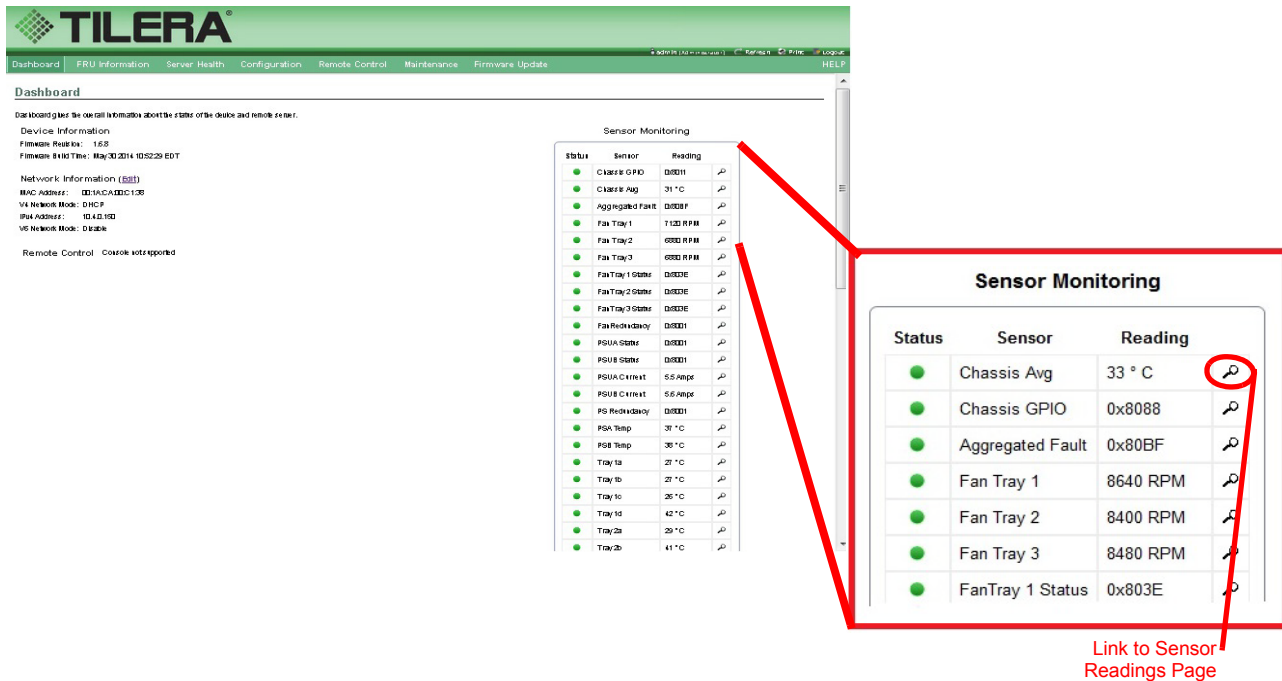


Figure 4-1: BMC Dashboard Sensor Monitoring Link

The **Sensor Readings** dialog appears, shown in Figure 4-2.

Note: When a customer views the **Dashboard** (or uses `ipmitool`) and sees “Not Available” for some sensors this is likely because that particular sensor is not available on that particular unit configuration. Platforms use different CPU/IO modules and can have different sensors available.

If the platform is powered down, ignore the sensor output. In some cases, the Dashboard displays “Not Available” or cached values from previous data (when the platform was powered on).

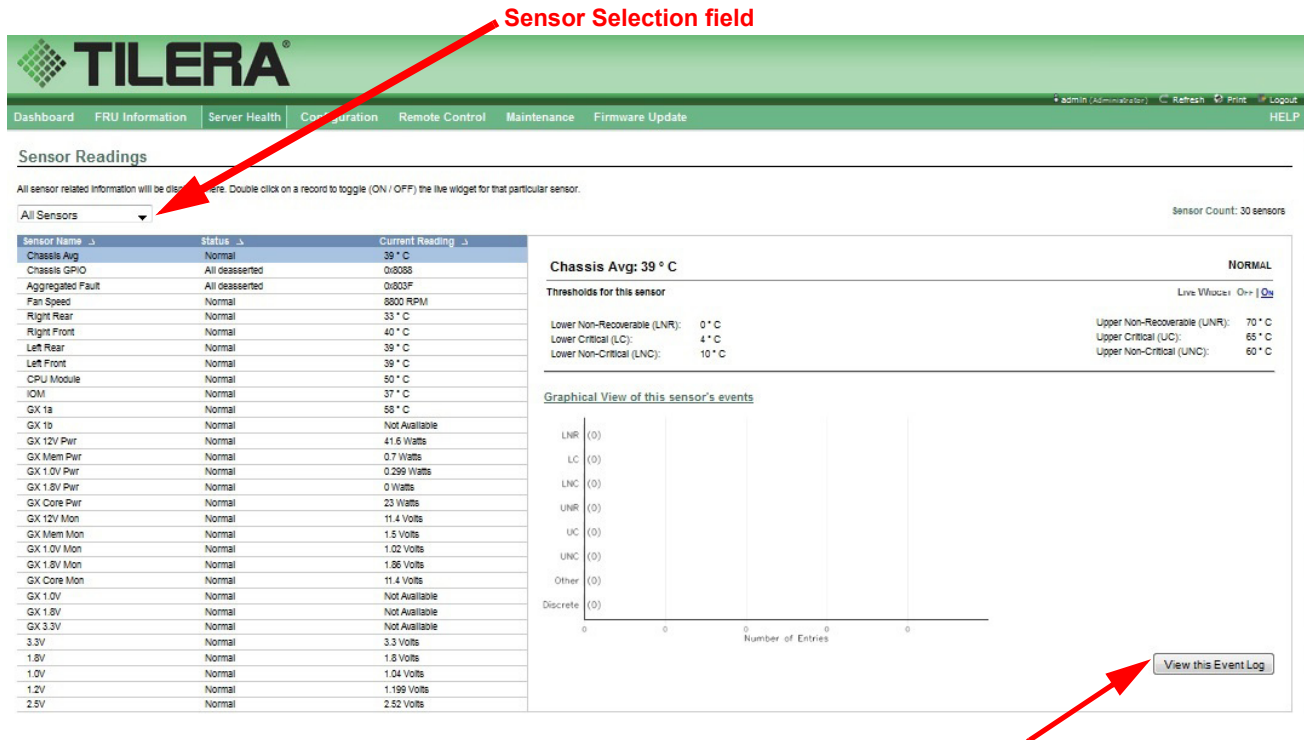


Figure 4-2: BMC Sensor Readings Screen

- Click on the sensor for which you want more data and then click **View this Event Log** to view the log of events for the sensor.

Note: You can view sensor data for a group by clicking on the **Sensor Selection** field and selection the group for which you want data. Options include:

- All Sensors
- Temperature Sensors
- Voltage Sensors
- Current Sensors
- Fan Sensors
- Power Supply
- Other Units-based Sensors
- Chassis

Event Log

Events generated by the system will be logged here. Double-click on a record to see description.

System Event Records

filter by: All Sensors

BMC Timezone

Client Timezone

UTC Offset: (GMT+/-0)

<<

<

1

>

>>

Event ID	Time Stamp	Sensor Name	Sensor Type	Description
6	04/30/2014 19:18:13	GX Mem Mon	Voltage	Lower Non-Recoverable - Going Low - Asserted
5	04/30/2014 19:18:13	GX Mem Mon	Voltage	Lower Critical - Going Low - Asserted
4	04/30/2014 19:18:13	GX Mem Mon	Voltage	Lower Non-Critical - Going Low - Asserted
3	04/30/2014 16:57:20	GX Mem Mon	Voltage	Lower Non-Recoverable - Going Low - Asserted
2	04/30/2014 16:57:20	GX Mem Mon	Voltage	Lower Critical - Going Low - Asserted
1	04/30/2014 16:57:20	GX Mem Mon	Voltage	Lower Non-Critical - Going Low - Asserted

Figure 4-3: BMC Event Log Screen

4.3 Viewing Sensor Alarm Thresholds

- From the **Dashboard** screen, select **Server Health**→**Sensor Readings** to display the **Sensor Readings** screen.
- Select the group of sensors for which you want data by clicking on the name of the sensor (Figure 4-4). Note that the list of sensors varies depending on the platform on which the BMC is installed.

Sensor Readings

All sensor related information will be displayed here. Double click on a record to toggle (ON / OFF) the live widget for that particular sensor.

Temperature Sensors

Sensor Count: 9 sensors

Sensor Name	Status	Current Reading
Chassis Avg	Normal	39 ° C
Right Rear	Normal	30 ° C
Right Front	Normal	38 ° C
Left Rear	Normal	40 ° C
Left Front	Normal	38 ° C
CPU Module	Normal	53 ° C
IOM	Normal	36 ° C
GX 1a	Normal	61 ° C
GX 1b	Normal	Not Available

GX 1a: 61 ° C

NORMAL

Thresholds for this sensor

Lower Non-Recoverable (LNR): 0 ° C

Lower Critical (LC): 4 ° C

Lower Non-Critical (LNC): 10 ° C

Upper Non-Recoverable (UNR): 105 ° C

Upper Critical (UC): 100 ° C

Upper Non-Critical (UNC): 95 ° C

Live Widget On | Off

Graphical View of this sensor's events

LNR (0)

LC (0)

LNC (0)

UNR (0)

UC (0)

UNC (0)

Other (0)

Discrete (0)

0000000

Number of Entries

View this Event Log

Figure 4-4: Sensor Readings Screen

- Double click on the sensor name to display a **Live Reading** of the sensor (see Figure 4-5). Note that you can double click on multiple sensors and have each displayed at the same time, each in its own view.

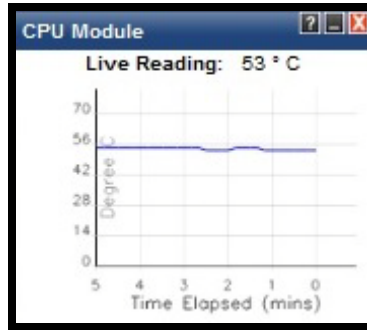


Figure 4-5: Live Reading Screen

4. Close a **Live Reading** screen by clicking on the **Close** icon.

4.4 Printing System Event Logs

1. To print System Event logs, first display the **Dashboard** screen and click on the **Magnifying glass** icon, as shown in [Figure 4-1](#).
2. Select a sensor name and click on **View this Event Log** to display the **Event Log** screen for the desired sensor.

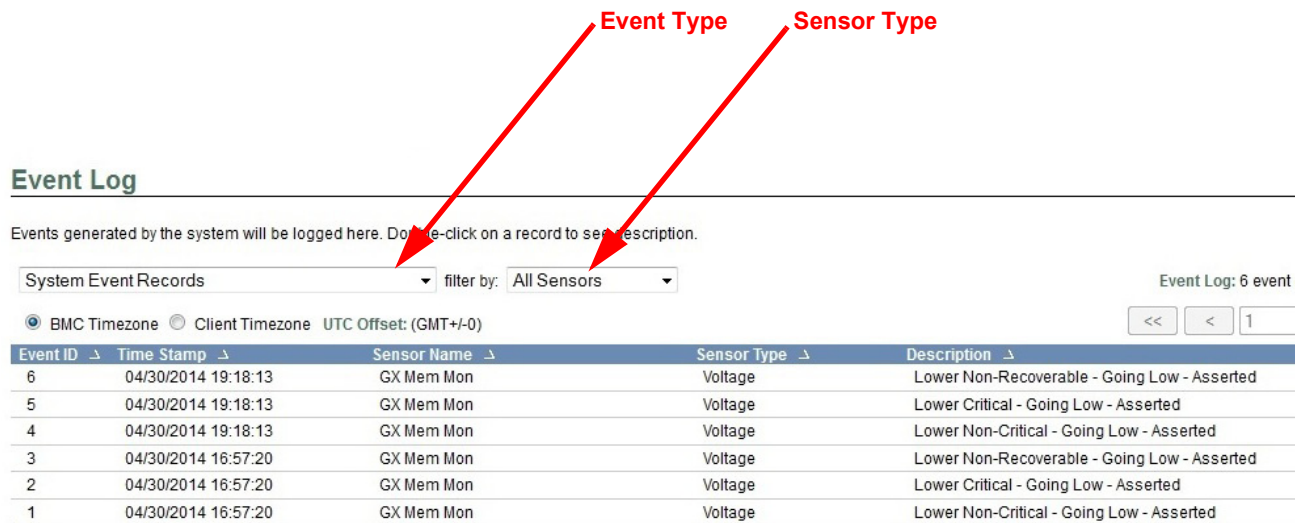


Figure 4-6: Event Log Screen

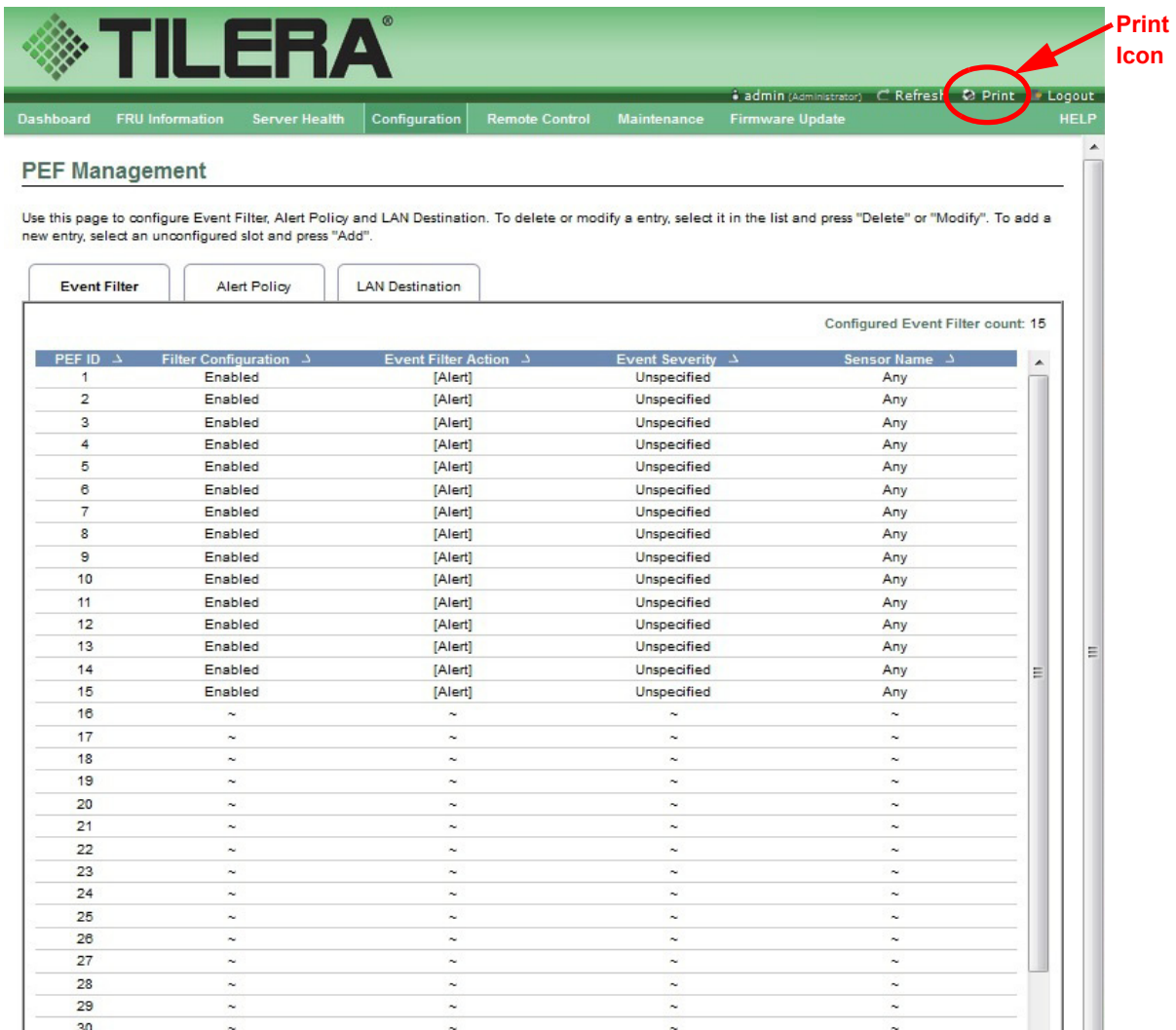
3. Select **Event Type** for which you want data. Options include:
 - All Events
 - System Event Records
 - OEM Event Records
 - BIOS Generated Events
 - SMI Handler Events
 - System Management Software Events

- System Software - OEM Events
 - Remote Console software Events
 - Terminal Mode Remote Console software Events
4. If the **Sensor Type** field is enabled, select the **Sensor Type**. Note that options vary based on Event Type selected.
 5. Click on the **Print** icon on the Tiler banner (see [Figure 4-7](#)).
The **Print** dialog is displayed with the default printer listed.
 6. Click **OK**.

4.5 Configuring Event Filters

This procedure describes how to configure event filters, known as Platform Event Filters (PEFs).

After logging on to BMC application, select **Configuration**→**PEF**.



TILERA

admin (Administrator) Refresh Print Logout

Dashboard FRU Information Server Health Configuration Remote Control Maintenance Firmware Update HELP

PEF Management

Use this page to configure Event Filter, Alert Policy and LAN Destination. To delete or modify a entry, select it in the list and press "Delete" or "Modify". To add a new entry, select an unconfigured slot and press "Add".

Event Filter Alert Policy LAN Destination

Configured Event Filter count: 15

PEF ID	Filter Configuration	Event Filter Action	Event Severity	Sensor Name
1	Enabled	[Alert]	Unspecified	Any
2	Enabled	[Alert]	Unspecified	Any
3	Enabled	[Alert]	Unspecified	Any
4	Enabled	[Alert]	Unspecified	Any
5	Enabled	[Alert]	Unspecified	Any
6	Enabled	[Alert]	Unspecified	Any
7	Enabled	[Alert]	Unspecified	Any
8	Enabled	[Alert]	Unspecified	Any
9	Enabled	[Alert]	Unspecified	Any
10	Enabled	[Alert]	Unspecified	Any
11	Enabled	[Alert]	Unspecified	Any
12	Enabled	[Alert]	Unspecified	Any
13	Enabled	[Alert]	Unspecified	Any
14	Enabled	[Alert]	Unspecified	Any
15	Enabled	[Alert]	Unspecified	Any
16	~	~	~	~
17	~	~	~	~
18	~	~	~	~
19	~	~	~	~
20	~	~	~	~
21	~	~	~	~
22	~	~	~	~
23	~	~	~	~
24	~	~	~	~
25	~	~	~	~
26	~	~	~	~
27	~	~	~	~
28	~	~	~	~
29	~	~	~	~
30	~	~	~	~

Figure 4-7: PEF Management Screen

CHAPTER 5 CONTROLLING POWER AND FANS

5.1 Overview

This chapter describes how to manage the power controls, fan speeds, and power to individual CPU and I/O modules from the BMC and by using ipmi tools.

Most procedures can be completed using Command Line Interface (CLI) or GUI-based methods, as listed in [Table 5-1](#).

Table 5-1. Common BMC IPMI Commands and GUI Equivalents

IPMI Function	Description	GUI Function
Section C.2.1 “Powering the Platform On/Off/Cycling or Resetting Using the CHASSIS POWER Command” on page 43	Powers the platform on/off/cycle or resets it.	Section 5.2.1 “Cycling Power on the Platform from the BMC” on page 30
Section C.2.2 “Causing the Chassis to Blink using the CHASSIS IDENTIFY Command” on page 43	Causes the chassis LEDs to blink for a specified period of time.	Not applicable
Section C.2.3 “Displaying Firmware Release Information Using the BMC INFO Command” on page 43	Gives basic information about the BMC.	Section 2.5 “Displaying BMC Release Information” on page 11
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Section C.2.9 “Managing User Accounts Using the USER Command” on page 47	Manages user accounts.	Section 2.4.3 “Managing User Accounts” on page 8

5.2 Managing Power Control from the BMC

You can use the BMC management interface to manage the power supply for the entire platform. You can perform the following power-related operations:

Reset Server	Reset all of the modules.
Power Off Server - Immediate	Shut down the platform.
Power On Server	Power up the platform after powering it off either by the front panel power switch or by either Power Off operations on the Power Control and Status screen.
Power Cycle Server	Shut down and then power on all devices in the platform.

5.2.1 Cycling Power on the Platform from the BMC

To turn on the platform from the BMC management interface, do the following:

1. Choose **Remote Control** → **Server Power Control**. The **Power Control and Status** screen appears.

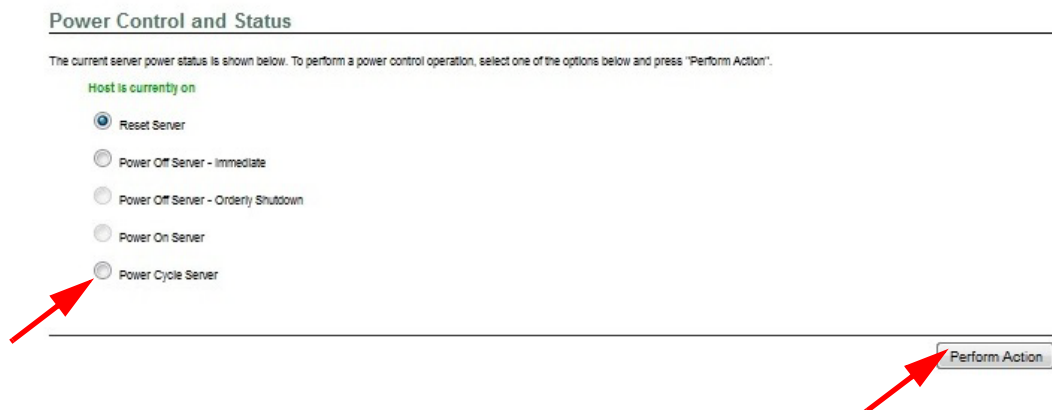


Figure 5-1: Cycling the Power on the Platform from the BMC

2. Choose the **Power Cycle Server** button.
3. Click **Perform Action**.

This causes the platform to power off and then power on again in sequence. While it is power cycling, the following message is displayed:

PERFORMING POWER ACTION...PLEASE WAIT ■■■

5.2.2 Resetting the Platform from the BMC

To turn on the platform from the BMC management interface, do the following:

1. From the **Dashboard**, choose **Remote Control** → **Server Power Control**. The **Power Control and Status** screen appears.

Power Control and Status

The current server power status is shown below. To perform a power control operation, select one of the options below and press "Perform Action".

Host is currently on

☒ Reset Server

☐ Power Off Server - Immediate

☐ Power Off Server - Orderly Shutdown

☐ Power On Server

☐ Power Cycle Server

Perform Action

Figure 5-2: Performing a Device Reset on the Platform from the BMC

2. Choose the **Reset Server** button.
3. Click **Perform Action**.

This operation sends a reset signal to each device inside the platform.

5.2.3 Turning Power Off for the Platform from the BMC

To turn power off on the platform from the BMC management interface, do the following:

1. From the **Dashboard**, choose **Remote Control** → **Server Power Control**. The **Power Control and Status** screen appears.

Power Control and Status

The current server power status is shown below. To perform a power control operation, select one of the options below and press "Perform Action".

Host is currently on

☐ Reset Server

☒ Power Off Server - Immediate

☐ Power Off Server - Orderly Shutdown

☐ Power On Server

☐ Power Cycle Server

Perform Action

Figure 5-3: Turning off the Platform from the BMC

2. Choose the **Power Off Server - Immediate** button.
3. Click **Perform Action**.

This is the equivalent of actuating the power switch on the front panel of the platform.

5.2.4 Turning on the Platform from the BMC

To turn on the platform from the BMC management interface, do the following:

1. From the **Dashboard**, choose **Remote Control** → **Server Power Control**. The **Power Control and Status** screen appears.

Power Control and Status

The current server power status is shown below. To perform a power control operation, select one of the options below and press "Perform Action".

Host is currently on

- ☐ Reset Server
- ☒ Power Off Server - Immediate
- ☐ Power Off Server - Orderly Shutdown
- ☐ Power On Server
- ☐ Power Cycle Server

Perform Action

Figure 5-4: Powering On the Platform from the BMC¹

2. Choose the **Power On Server** button.
3. Click **Perform Action**.

This is the equivalent of actuating the power switch on the front panel of the platform.

5.3 Managing Fan Speed Using the BMC

Depending on the operating environment in which the platform placed, it might be desirable to throttle fan speeds higher or lower. This command adjusts the base duty cycle used in fan speed control. Thus, depending on environmental conditions, a setting of 50% might cause fans to reach their maximum allowed speed.

You can set the fan speed as a percentage of that RPM range using `ipmitool raw` commands. Here are some examples:

- To set the fan speed to 25%, use the following command:

```
-bash-3.2$ ipmitool -H <hostname/IP> -U admin -P admin raw 0x38 0x80 25
```
- To set the fan speed to 100%, use the following command:

```
-bash-3.2$ ipmitool -H <hostname/IP> -U admin -P admin raw 0x38 0x80 100
```

Note: The fan speed control operations are slightly delayed. It might take a few minutes for the fan speed to change from maximum speed down to the lowest speed.

1. Note that this illustration shows that the host is currently on and the Power On Server function disabled. If the server is off, this option is enabled.

5.4 Controlling Power for Individual CPU and I/O Modules

From your Linux host machine, you can use the `ipmitool` command to power on/off individual CPU or IO Modules or take them in or out of reset. The minimum version required is `ipmitool-1.8`.

Note: You must have the `ipmitool` binary installed on your Linux x86 host system to use the `ipmitool` command in this way. Run the `rpm -i` or `yum` commands to install the RPM package on your host machine.

The syntax for power management commands using IPMI on the BMC is as follows:

```
ipmitool -H bmc_ip_addr -U username -P password raw net_func command [subcommand mask]
```

The parameters for the `ipmitool` command are:

<i>bmc_ip_addr</i>	The IP address for the BMC interface.	
<i>username</i>	A valid username to logon to the BMC; e.g. admin.	
<i>password</i>	The password for the username to logon to BMC; e.g. admin.	
<i>net_func</i>	The value must always be 0x38 (for Tiler OEM commands).	
<i>command</i>	An IPMI command number specified in hex notation, which could be:	
	0x0	Get Module Status. This command does not have sub-commands.
	0x1	Set Module Status. This command has sub-commands.
	0x2	Set the delay for the next power-off.

Note: The `Set Delay` command (0x2) applies to the entire chassis, so powering off a single module is also affected by this delay. Note also that this delay is “sticky” so if you want a 0-delay, you must explicitly re-issue the command with a 0 delay.

<i>subcommand</i>	An IPMI sub-command number specified in hex notation, which could be:	
	0x0	Set Module Power
	0x1	Set Module Reset
	0x2	Cycle Module Power
	0x3	Cycle Module Reset
<i>mask</i>	See Section 5.4.1 Understanding the Bit Mask for an IPMI Raw Command .	

Note: Tiler® recommends that you always control power or reset modes for CPU modules and I/O modules in tandem—if you cycle power for CPU module #2, also cycle power for the associated I/O module #2.

5.4.1 Understanding the Bit Mask for an IPMI Raw Command

You can perform IPMI control operations upon the four CPU modules and I/O modules independently, by creating a bit-mask to identify the devices upon which you want to have an effect using a particular sub-command.

Build a mask by setting the appropriate bits of the byte that you supply to the `ipmitool` command as the last parameter (the *mask*), according to the specification in [Figure 5-5](#). This mask is typically specified as a hex value, but a decimal value works as well.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
I/O Module #4	I/O Module #3	I/O Module #2	I/O Module #1	CPU Module #4	CPU Module #3	CPU Module #2	CPU Module #1

Figure 5-5: Correspondence Between Device and Bit Mask Bit Location

For some control commands, you set the bit for a particular device to 1 and for other control commands you set the bit to 0, as follows:

- To power on a specific device, set its corresponding binary bit to 1 in the mask. Set the bit to 0 for a device that you want to have no power.
Set Module Power (sub-command 0x0)
- To reset specific device, set its corresponding binary bit to 0 in the mask. To leave a device unchanged, set the bit to 1.
Set Module Reset (sub-command 0x1)
- To power-cycle a specific device, set its corresponding binary bit to 1 in the mask. To leave a device unchanged, set the bit to 0.
Cycle Module Power (sub-command 0x2)
- To place a device in reset and then out again, set its corresponding bit to 1 in the mask. To leave a device unchanged, set the bit to 0.
Cycle Module Reset (sub-command 0x3)

Some mask examples are:

- To power on all CPU modules but no I/O modules, set the mask to 0b00001111, which translates to the hexadecimal code 0x0F.
- To turn power off only for CPU module #2 and I/O module #2 and leave power all for all other devices, the mask is 0b11011101, which translates to the Hexadecimal value 0xDD.
- To cycle power for only CPU #3 and I/O module #3, the mask is 0b01000100, which is 0x44.

5.4.2 Getting Power Status Using an ipmitool Raw Command

The BMC responds to an ipmitool raw Get Module Status request by sending a pair of response bytes, indicating those devices which are powered and those devices which are in reset state.

To query that status of the CPU and I/O modules in the device at 10.8.0.150, use this command (in hex but without the “0x” prefix):

```
-bash-3.2$ ipmitool -H 10.8.0.150 -U admin -P admin raw 0x38 0x0
33 ff
```

In this example, the first response byte indicates that power is on for CPU modules #1 and #2 and for I/O modules #1 and #2, 33, with power being off for all other devices. The second bytes indicates that no devices are in reset.

Note: You do not need to supply either a sub-command code or a mask for a Get Module Status command.

5.4.3 Turning Power Off for All Modules using ipmitool

To turn off power to all of the CPU modules and I/O modules in the device at 10.8.0.150, use this command:

```
-bash-3.2$ ipmitool -H 10.8.0.150 -U admin -P admin raw 0x38 0x1 0x0 0x00
```

Note: You can achieve the same affect by using the **Power Off Server - Immediate** button on the **Power Control and Status** screen in the BMC management interface.

5.4.4 Turning Power On for All Modules using ipmitool

To turn power on for all of the CPU modules and I/O modules in the device at 10.8.0.150, use this command:

```
-bash-3.2$ ipmitool -H 10.8.0.150 -U admin -P admin raw 0x38 0x1 0x0 0xFF
```

Note: You can achieve the same affect as this using the **Power On Server** button on the **Power Control and Status** screen in the BMC management interface.

5.4.5 Cycling Power to Some Modules Using ipmitool

To cycle power on CPU and I/O modules #2, #3 and #4 in the device at 10.8.0.150, issue this command:

```
-bash-3.2$ ipmitool -H 10.8.0.150 -U admin raw 0x38 0x1 0x2 0xEE
```

5.4.5.1 Resetting a CPU Module Using ipmitool

To place only the TILE-Gx™ processor on CPU module #4 (and the associated I/O module #4) into reset, issue the following `ipmitool` RAW command:

```
-bash-3.2$ ipmitool -H hostname/IP -U admin -P admin raw 0x38 0x1 0x1 0x77
```

The BMC sends a reset signal to CPU module #4 and I/O module #4 and leaves all of the other CPU and I/O modules unchanged.

5.4.5.2 Cycling Reset for Some Modules Using ipmitool

To cycle reset for CPU and I/O modules #1 and #4 in the device at 10.8.0.150, issue this command:

```
-bash-3.2$ ipmitool -H 10.8.0.150 -U admin -P admin raw 0x38 0x1 0x3 0x99
```

APPENDIX A: REFERENCES AND ADDITIONAL BMC MATERIALS

A.1 IPMI Web Pages

IPMITool Homepage

<http://ipmitool.sourceforge.net>

Intelligent Platform Management Interface Specification

<http://www.intel.com/design/servers/ipmi>

FreeIPMI Homepage

<http://www.gnu.org/software/freeipmi/>

A.2 man Pages

If you have the utility installed, the IPMITool supports man pages for `ipmitool`. Refer to the IPMITool homepage listed above.

APPENDIX B: FLASH UTILITIES: USING ALTERNATE METHODS FOR UPGRADING BMC SOFTWARE

B.1 Overview

This appendix describes the YAFUFlash utility and explains how to use it to “flash” the BMC.

Note: Yafuflash works with BMC Release 1.6.8 or later. Yafuflash does not work with BMC Rel. 1.6.6.

B.2 Using Yafuflash

1. On an x86 Linux host, change directories (cd) to the directory containing the Yafuflash binary.
2. Run Yafuflash in the terminal. The syntax is as follows:

```
./Yafuflash [OPTION] [MEDIUM] <FW_IMAGE_FILE>
```

Options	Description
-info	This option displays information relating to existing and current firmware.
-auto	This option allows for automatic upgrades. It begins by comparing flash module headers.
-full	This option enables full upgrades.
-force-boot	This option forces the boot loader to be upgraded during a full upgrade.
-c	This option preserves configuration settings during the full upgrade.

Medium Options	Description
-cd	This option enables USB connections.
-nw -ip	This option enables network connections using -ip (followed by the IP address).

Examples

The following examples, the firmware image filename <FW_IMAGE_FILE> is rom.ima and the [MEDIUM] is -nw (network):

- To display the details of both existing and new firmware using the network connection with the IP address is 155.166.132.12:

```
./Yafuflash -nw -ip 155.166.132.12 -info rom.ima
```

- To start flashing the new rom.ima to the firmware using the network connection with the IP address of 155.166.132.12:

```
./Yafuflash -nw -ip 155.166.132.12 -full rom.ima
```

- To start flashing the new rom.ima to the firmware with the FORCE BootLoader Upgrade via the network connection using the IP address of 155.166.132.12.

```
./Yafuflash -nw -ip 155.166.132.12 -full -force-boot rom.ima
```

APPENDIX C: COMMON BMC IPMI COMMANDS AND GUI EQUIVALENTS

This appendix describes some commonly available `ipmitool` commands that Tiler® supports. Parameters shown in *italics* represent variables.

C.1 ipmitool(1) Syntax

For additional information, refer to the `ipmitool` man page for reference at:

```
ipmitool [-c|-h|-d N|-v|-V] -I open <command>
```

```
ipmitool [-c|-h|-v|-V] -I lan -H <hostname> [-p <port>]  
[-U <username>]  
[-A <authtype>]  
[-L <privlvl>]  
[-a|-E|-P|-f <password>]  
[-o <oemtype>]  
[-O <sel oem>]  
[-e <esc_char>]  
[-N <sec>]  
[-R <count>]  
<command>
```

```
ipmitool [-c|-h|-v|-V] -I lanplus -H <hostname> [-p <port>]  
[-U <username>]  
[-L <privlvl>]  
[-a|-E|-P|-f <password>]  
[-o <oemtype>]  
[-O <sel oem>]  
[-C <ciphersuite>]  
[-Y|[-K|-k <kg_key>]  
[-y <hex_kg_key>]  
[-e <esc_char>]  
[-N <sec>]  
[-R <count>]  
<command>
```

<http://linux.die.net/man/1/ipmitool>

C.2 Common BMC IPMI Commands and GUI Equivalents

Most procedures can be completed using Command Line Interface (CLI) or GUI-based methods, as listed in [Table C-1](#).

Table C-1. Common BMC IPMI Commands and GUI Equivalents

IPMI Function	Description	GUI Function
Section C.2.1 “Powering the Platform On/Off/Cycling or Resetting Using the CHASSIS POWER Command” on page 43	Powers the platform on/off/cycle or resets it.	Section 5.2.1 “Cycling Power on the Platform from the BMC” on page 30
Section C.2.2 “Causing the Chassis to Blink using the CHASSIS IDENTIFY Command” on page 43	Causes the chassis LEDs to blink for a specified period of time.	Not applicable
Section C.2.3 “Displaying Firmware Release Information Using the BMC INFO Command” on page 43	Gives basic information about the BMC.	Section 2.5 “Displaying BMC Release Information” on page 11
Section C.2.4 “Changing the Network Configuration Using the LAN Command” on page 44	Changes the network configuration for eth0 and eth1 (internal network if available).	Section 3.4 “Changing the Network Configuration” on page 20
Section C.2.5 “Viewing all the IPMI Sensors Using the SDR Command” on page 44	Views all IPMI Sensors.	Section 4.2 “Viewing Sensor Data Captured by the BMC” on page 24
Section C.2.5 “Viewing all the IPMI Sensors Using the SDR Command” on page 44	Views detailed information on IPMI Sensors and change sensor alarm thresholds.	Section 4.3 “Viewing Sensor Alarm Thresholds” on page 26
Section C.2.7 “Printing System Event Logs Using the SEL Command” on page 47	Prints the System Event Log.	Section 4.4 “Printing System Event Logs” on page 27
Section C.2.8 “Configuring Event Filters Using the PEF Command” on page 47	Configures event filters, alert policies, and LAN destinations.	Section 4.5 “Configuring Event Filters” on page 28
Section C.2.9 “Managing User Accounts Using the USER Command” on page 47	Manages user accounts.	Section 2.4.3 “Managing User Accounts” on page 8

C.2.1 Powering the Platform On/Off/Cycling or Resetting Using the CHASSIS POWER Command

Use this command to power the platform on/off/cycle or reset it. Once a system is powered off, only BMCs installed in platforms that support Standby power can still be accessed (TILEmpower-Gx™ and TILEmpower-Gx FR only).

This functionality is also displayed through the Web GUI.

Example

```
ipmitool -U admin -P admin -H <hostname or IP Address> chassis power
```

Returns

chassis power Commands: status, on, off, cycle, reset

C.2.2 Causing the Chassis to Blink using the CHASSIS IDENTIFY Command

This command causes the Chassis LEDs to blink for specified period of time. Support for this command is available on platforms for which the BMC controls the LEDs.

Example

```
ipmitool -U admin -P admin -H <hostname or IP Address> chassis identify <seconds>
```

C.2.3 Displaying Firmware Release Information Using the BMC INFO Command

This command gives basic information about the BMC. Currently, it is only useful to display the firmware revision. Field Replaceable Unit (FRU) information is not yet supported.

The firmware revision is also displayed on the Web GUI [Dashboard](#).

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> bmc info
```

Returns

```
Device ID : 32
Device Revision : 1
Firmware Revision : 1.6
IPMI Version : 2.0
Manufacturer ID : 0
Manufacturer Name : Unknown
Product ID : 43707 (0xaabb)
Product Name : Unknown (0xAABB)
Device Available : yes
Provides Device SDRs : no
Additional Device Support :
    Sensor Device
    SDR Repository Device
    SEL Device
    FRU Inventory Device
    IPMB Event Receiver
    IPMB Event Generator
    Chassis Device
```

```
Aux Firmware Rev Info      :
0x08
0x00
0x00
0x00
```

C.2.4 Changing the Network Configuration Using the LAN Command

This command can change the network configuration for eth0 and eth1 on the internal network, if they are available.

This functionality is also available through the [Web GUI→Configuration, Network](#).

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> lan
```

Returns

```
LAN Commands:
print [<channel number>]
set <channel number> <command> <parameter>
alert print <channel number> <alert destination>
alert set <channel number> <alert destination> <command> <parameter>
stats get [<channel number>]
stats clear [<channel number>]
```

C.2.5 Viewing all the IPMI Sensors Using the SDR Command

View all IPMI Sensors or by group:

These values are also displayed on the Web GUI [Dashboard](#).

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> sdr
```

Returns

Aggregated Fault	0x3f	ok
Chassis GPIO	0x88	ok
FanTray 1 Status	0x3e	ok
FanTray 2 Status	0x3e	ok
FanTray 3 Status	0x3e	ok
FanRedundancy	0x01	ok
PSUA Status	0x01	ok
PSUB Status	0x01	ok
PS Redundancy	0x01	ok
PSUA Current	2.60 Amps	ok
...		

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> sdr help
```

Returns

```
SDR Commands:  list | elist [all|full|compact|event|mcloc|fru|generic]
                all      All SDR Records
                full      Full Sensor Record
                compact    Compact Sensor Record
                event      Event-Only Sensor Record
                mcloc      Management Controller Locator Record
                fru        FRU Locator Record
                generic     Generic Device Locator Record
type [sensor type]
list           Get a list of available sensor types
get           Retrieve the state of a specified sensor
info          Display information about the repository itself
entity <id>[.<instance>]
              Display all sensors associated with an entity
dump <file>    Dump raw SDR data to a file
fill
sensors       Creates the SDR repository for the current configuration
file <file>   Load SDR repository from a file
```

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> sdr type
voltage
```

Returns

GX 12V Mon	24h	ok	7.1	11.60 Volts
GX Mem Mon	25h	ok	7.1	1.50 Volts
GX 1.0V Mon	26h	ok	7.1	1 Volts
GX 1.8V Mon	27h	ok	7.1	1.86 Volts
GX Core Mon	28h	ok	7.1	11.70 Volts
GX 1.0V	29h	ok	7.1	2.34 Volts
GX 1.8V	2Ah	ok	7.1	2.34 Volts
GX 3.3V	2Bh	ok	7.1	2.34 Volts
3.3V Standby	2Ch	ok	7.1	3.32 Volts
1.8V Standby	2Dh	ok	7.1	1.79 Volts
1.26V Standby	2Eh	ok	7.1	1.25 Volts

C.2.6 Viewing Detailed Information on Sensors and Changing Alarm Thresholds Using the SENSOR Command

Use this command to view detailed information on IPMI sensors and change sensor alarm thresholds.

These values are also available through the [Web GUI → Server Health, Sensor Readings](#).

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> sensor help
```

Returns

```
Sensor Commands:  list thresh get reading
```

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> sensor get "GX 1a"
```

Returns

```
Locating sensor record...
Sensor ID          : GX 1a (0x1d)
Entity ID          : 7.1
Sensor Type (Analog) : Temperature
Sensor Reading      : 57 (+/- 0) degrees C
Status             : ok
Lower Non-Recoverable : -8.000
Lower Critical      : -4.000
Lower Non-Critical   : -1.000
Upper Non-Critical   : 95.000
Upper Critical       : 100.000
Upper Non-Recoverable : 108.000
Assertion Events     :
  Assertions Enabled  : lnc- lcr- lnr- unc+ ucr+ unr+
  Deassertions Enabled : lnc- lcr- lnr- unc+ ucr+ unr+
```

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> sensor
```

Returns

Aggregated Fault	0x0	discrete	0x3f80	na	na	na	na	na	na
Chassis GPIO	0x0	discrete	0x8880	na	na	na	na	na	na
FanTray 1 Status	0x0	discrete	0x3e80	na	na	na	na	na	na
FanTray 2 Status	0x0	discrete	0x3e80	na	na	na	na	na	na
FanTray 3 Status	0x0	discrete	0x3e80	na	na	na	na	na	na
FanRedundancy	0x0	discrete	0x0180	na	na	na	na	na	na
PSUA Status	0x0	discrete	0x0180	na	na	na	na	na	na
PSUB Status	0x0	discrete	0x0180	na	na	na	na	na	na
PS Redundancy	0x0	discrete	0x0180	na	na	na	na	na	na
PSUA Current	2.600	Amps	ok	na	na	na	na	na	na
PSUB Current	5.000	Amps	ok	na	na	na	na	na	na
Fan Tray 1	8800.000	RPM	ok	240.000	480.000	800.000	16000.000	16640.000	17280.000
Fan Tray 2	8720.000	RPM	ok	240.000	480.000	800.000	16000.000	16640.000	17280.000

C.2.7 Printing System Event Logs Using the SEL Command

Use this command to print the System Event Log.

This functionality is also available through the [Web GUI](#) → [Server Health, Event Log](#).

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> sel
```

Returns

```
SEL Information
Version          : 1.5 (v1.5, v2 compliant)
Entries          : 0
Free Space       : 65502 bytes
Percent Used     : 0%
Last Add Time    : Not Available
Last Del Time    : Not Available
Overflow         : false
Supported Cmds   : 'Delete' 'Partial Add' 'Reserve' 'Get Alloc Info'
# of Alloc Units : 3639
Alloc Unit Size  : 18
# Free Units     : 3639
Largest Free Blk : 3639
Max Record Size  : 14
```

C.2.8 Configuring Event Filters Using the PEF Command

Use this command to configure event filters, alert policies, and LAN destinations. Refer to the IPMI 2.0 specification for more details:

<http://www.intel.com/content/www/us/en/servers/ipmi/ipmi-specifications.html>

This functionality is also available through the [Web GUI](#) → [Configuration, PEF](#).

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> pef help
```

C.2.9 Managing User Accounts Using the USER Command

Use this command to manage user accounts.

This functionality is also available through the [Web GUI](#) → [Configuration, Users](#).

Example

```
-bash-3.2$ ipmitool -U admin -P admin -H <hostname or IP Address> user
```

Returns

```
User Commands: summary [<channel number>]
list            [<channel number>]
set name        <user id> <username>
set password    <user id> [<password>]
disable         <user id>
enable         <user id>
priv            <user id> <privilege level> [<channel number>]
test            <user id> <16|20> [<password>]
```

C.3 Other BMC Web GUI Features

Note: When a customer views the **Dashboard** (or uses the `ipmitool`) and sees “Not Available” for some sensors, this is because that particular sensor is not available on that particular unit configuration. Platforms use different CPU/IO Modules and may have different sensors available.

If the platform is powered down, the sensor output should be ignored as it can display “Not Available” or cached values retained when the platform was powered on.

Table C-2. Other BMC Web GUI Features

Sub-Function	IPMI Commands	Access to Data via GUI ¹
Sensor Monitoring		
--	SDR	Dashboard option
Device Information		
	BMC INFO	
Firmware Revision (number)		Dashboard option. The Firmware Revision number is displayed under the Device Information heading.
Firmware Build Time		
Event Log Pie Chart		
	SEL or PEF	Dashboard option
FRU Information		
		FRU Information option Currently not supported.
Server Health		
Sensor Readings	SENSOR	Server Health→Sensor Readings
Event Log	• SEL to display and delete • PEF to setup filtering	Server Health→Event Log
System and Audit Log		Server Health→System and Audit Log
Configuration		
Active Directory		Configuration→Active Directory
DNS (see AMI OEM Commands)		Configuration→DNS
Event Log (Circular or Linear Buffers)		Configuration→Event Log
LDAP/E Directory Settings (Light-weight Directory Access Protocol)		Configuration→LDAP/E-Directory Settings Refer to: http://en.wikipedia.org/wiki/Light-weight_Directory_Access_Protocol

Table C-2. Other BMC Web GUI Features (continued)

Sub-Function	IPMI Commands	Access to Data via GUI ¹
Network	LAN	Configuration→Network Settings
NTP		Configuration→NTP Settings
PAM Ordering (AMI OEM command)		Configuration→PAM Ordering Default PAM and NSS order for Set and Get Command.
	PEF	event filter
	RADIUS	Configuration→RADIUM RADIUS authentication server configuration.
SSL		Maintenance→Preserve Configuration
System and Audit Log AMI OEM commands: • Audit Log Configuration • System Log Configuration		
Users AMI OEM commands: • Get Email Address of the USER • Set Email Address of the USER	USER	
Remote Control		
Server Power Control	CHASSIS POWER	Remote Control→Server Power Control
Maintenance		
Preserve Configuration		Configuration→SSL NOTE: Not all configuration settings are maintained. SSH is guaranteed to be preserved, others are not.
Restore Factory Defaults AMI OEM Command: Default PAM and NSS order for Set and Get Command		Maintenance→Restore Factory Defaults
System Administrator		Maintenance→System Administrator
Firmware Update		
Firmware Update		Firmware Update→Firmware Update
Protocol Configuration		Firmware Update→Protocol Configuration

1. All GUI-based access points assume that you have logged onto BMC.

Appendix D: Supported OEM Commands

This appendix presents the commands, listed in [Table D-1](#), common to all Intelligent Platform Management (IPM) devices that support this specification's message/command interface.

Note: In this implementation, there is no support for IPv6-related commands.

Table D-1. AMI OEM Commands

Term	Definition
Firewall Configuration Commands	• Set FIREWALL Configuration Command • Get FIREWALL Configuration Command
Reset Password and User Email Commands	• Get Email Address of the USER • Set Email Address of the USER • Reset Password Command
Restore Default Commands	• Restore Default Configuration
Login Audit Commands	• Set Login Audit Configuration • Get Login Audit Configuration
Get/Set SEL Policy Commands	• Get SEL Policy Command • Set SEL Policy Command

D.1 Firewall Configuration Commands

The `FIREWALL` command is used to add/remove the `IPTABLES` rules in BMC to Restrict/Accept the communication from the client machines. `IPTABLES`' rules can be used to accept/drop the packets from client machines. We have set of rules to accept/drop the packets at different level of transmission of the packets (TCP/IP). You can restrict the IP Address and port number by adding the `IPTABLES`' rules. `DROP` rules block the IP Address or Port number and the `ACCEPT` rule unblocks the IP Address or Port number. The `Set FIREWALL` command is used to add/remove these rules in the Baseboard Management Controller (BMC).

Commands include:

`Set FIREWALL`. `Set FIREWALL` command is used to add/remove the `IPTABLES` rules to accept/drop the packets.

- `DROP` (packets). This state is used to drop the packets based on the rules.
- `ACCEPT` (packets). This state is used to accept the packets from the clients.

D.1.1 Set FIREWALL Configuration Command

The Set FIREWALL command is used to add/remove the IPTABLES rules, which allows packets to be accepted or dropped. Parameter 0-3 is used to add the IPTABLES rules and 4-7 is used to remove the IPTABLES rules that are added by parameter 0-3.

NetFn ¹	0x32
Command	0x76

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Data Field
1	Parameter Selector
2	State Selector
3:N	Configuration Parameter Data

Response Data

Byte	Completion Code	Description
1	00h	Success
	C7h	Invalid request length
	80h	Parameter not supported
	82h	Attempt to write read-only parameter

Table D-2. Set FIREWALL Configuration Parameter Details

Number	Parameter	Parameter Data
0	Add the rule for an IPv4 address	Data 1:4 IP Address MS-byte first. This is an IPv4 type Address, which is to be blocked or unblocked based on the state.
1	Add the rule for range of IPv4 addresses	Data 1:8 IP Address Range [1:4] Starting IP Address, from which IPs are to be blocked or unblocked based on the state. [5:8] Ending IP Address, till which IPs are to be blocked or unblocked based on the state.

Table D-2. Set FIREWALL Configuration Parameter Details (continued)

Number	Parameter	Parameter Data
2	Add the rule for the port number	Data 1 Protocol TCP/UDP 0 - TCP 1 - UDP 2 - Both TCP/UDP Data 2:3 Port No [2:3] Port No which is to be blocked or unblocked based on the state.
3	Add the rule for the port range	Data 1 Protocol TCP/UDP 0 - TCP 1 - UDP 2 - Both TCP/UDP Data 2:5 - Port Range [2:3] Port number from which Ports are to be blocked or unblocked based on the state. [4:5] Port number till which Ports are to be blocked or unblocked based on the state.
4	Remove the rule for the IPv4 Address	Data 1:4 IP Address MS-byte first. This is an IPv4 type Address, that are to be blocked or Unblocked based on the state.
5	Remove the rule for range of IPv4 addresses	Data 1:8 IP Address Range [1:4] Starting IP Address, from which IPs are to be blocked or Unblocked based on the state. [5:8] Ending IP Address, till which IPs are to be blocked or Unblocked based on the state.
6	Remove the rule for port number	Data 1 Protocol TCP/UDP 0 - TCP 1 - UDP 2 - Both TCP/UDP Data 2:3 - Port No [2:3] Port No which is blocked or Unblocked based on the state.
7	Remove the rule for port range	Data 1: - Protocol TCP/UDP 0 - TCP 1 - UDP 2 - Both TCP/UDP Data 2:5 Port Range [2:3] Port number from which Ports are to be blocked or Unblocked based on the state. [4:5] Port number till which Ports are to be blocked or Unblocked based on the state.
8	Flush iptable	Flush all the rules set using iptables
9	Drop all	Block all the ports and IPs from access
a	Remove Drop all	Remove drop all command

Set firewall configuration state details.

Table D-3. Firewall Configuration State Details

#	Type Specific Parameter	Parameter Data
00	To set the Command to DROP	Parameter to drop packets. This state can be used by parameter 0-3 to add the rules to drop the packets based on (IP address/Port number or range of IP address/Port number). Parameter 4-7 is used to remove the rule.
01	To set the Command to ACCEPT	Parameter to accept packets. This state can be used by parameter 0-3 to add the rules to accept the packets based on (IP address/Port number or range of IP address/Port number). Parameter 4-7 is used to remove the rule.

Examples

- To add the rule to drop the packets from client 10.0.6.130:

```
ipmitool -H 10.0.6.130 -U admin -P admin raw 0x32 0x76 0x0 0x0 0xa 0x0 0x06 0x82
```

Note that the IP address in hex would be 0x0A.00.06.82.

- To add the rule to accept the packets from client 10.0.6.130:

```
ipmitool -H <hostname or IP Address> -U admin -P admin raw 0x32 0x76 0x0 0x1 0xa 0x0 0x06 0x82
```

- To delete the rule to drop the packets from client 10.0.6.130:

```
ipmitool -H 10.0.6.130 -U admin -P admin raw 0x32 0x76 0x04 0x0 0xa 0x0 0x06 0x82
```

Note that the rules are processed in order. So in this example, client 10.0.6.130 cannot access the BMC, since the DROP rule already took effect before the ACCEPT rule. (And if the order is reversed, client 10.0.6.130 is always be ACCEPTed.)

D.1.2 Get FIREWALL Configuration Command

This Command is used to return the Firewall rules set to IPMI firmware.

NetFn ¹	0x32
Command	0x77

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tlera's set of command functions use 0x38.

Request Data

Byte	Data Field
1	Parameter Selector
2	Entry No Selector

Response Data

Byte	Completion Code	Description
1	00h	Success
	CCh	This is an invalid data field.
	80h	Parameter is not supported.
2	TotalCount / TYPE	Depends on the parameter selected.
3:N		Returns the firewall data set.

Table D-4. Get FIREWALL Configuration Parameter Details

Number	Parameter	Parameter Data
0	Get the number of firewall rules	The command returns the number of Firewall rules set using IPTABLES. If you enter the '0x0' parameter, you do not need to pass Entry number.
1	Get the details of a firewall rule	The command returns the details of a firewall rule set, for example: If a rule set is to block an IP, it returns the IP blocked with type as: =>0x0 <IP >

The following are the get firewall configuration entry number details:

EntryNo	This byte is the firewall rule number.
---------	--

Examples

01	First rule
05	Fifth rule

D.2 Reset Password and User Email Commands

D.2.1 Get Email Address of the USER

This command is used to get the email address of the user.

NetFn ¹	0x32
Command	0x63

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Completion Code	Description
1	userId	User identification number.

Response Data

Byte	Completion Code	Description
1	00h	Success
	CCh	Invalid Data Field
	C7h	Requested data length is invalid.
64	Email Address	Email Address

D.2.2 Set Email Address of the USER

This command is used to set the email address of the user.

NetFn¹ 0x32

Command 0x64

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Completion Code	Description
1	userId	User identification number.
64	Email Address	Email Address

Response Data

Byte	Completion Code	Description
1	00h	Success
	CCh	Invalid Data Field
	C7h	Requested data length is invalid.

D.2.3 Reset Password Command

This command is used to reset the password of the user.

NetFn ¹	0x32
Command	0x65

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Completion Code	Description
1	userId	User identification number.
1	Channel	Channel

Response Data

Byte	Completion Code	Description
1	00h	Success
	CCh	Invalid Data Field
	10h	Email not configured
	11h	SMTP disabled
	12h	Unsupported authentication type
	13h	Email authentication failed
	14h	Unable to connect to SMTP server
	15h	Email send Failed

D.3 Restore Default Commands

D.3.1 Restore Default Configuration

This command is used to restore the default configurations values.

NetFn ¹	0x32
Command	0x66

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Not applicable

Response Data

Byte	Completion Code	Description
1	00h	Success
	D1h	Failed due to Firmware in Update Mode
	FFh	Unspecified Error

D.3.2 Set Linux Root User Password Command

This command is used to set Linux root user password and to enable and disable.

NetFn ¹	0x32
Command	0x91

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Completion Code	Description
1	Operation	[7:2] Reserved
		[1:0] Operation 00b Disable user 01b Enable user 10b Set password
2:64	Password data	Password size should be a minimum of 8 bytes and maximum 64 bytes. This field need not be present if the operation is either to 'disable user' or 'enable user'. If this field is present for those operations, the BMC ignores the data.

Response Data

Byte	Completion Code	Description
1	00h	Success
	CCh	Invalid Data Field
	C6h	Invalid Request Length
	C1h	Invalid Command

D.4 Login Audit Commands

D.4.1 Set Login Audit Configuration

This command is used to configure login Audit events like enabling/disabling the System Event Logs for RMCP/Web user login/logouts.

NetFn¹ 0x32

Command 0x97

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Data Field	Description
1	Web Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable
2	RMCP/RMCP+ Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable

Byte	Data Field	Description
3	Telnet Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable
4	SSH Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable

Response Data

Byte	Completion Code	Description
1	00h	Success

Note: When you enable Audit configuration for the above mentioned services, you add the OEM-specific SEL record (record type - c1) for the each enabled event. SEL record bytes details are mentioned below. This record is specific for AMI.

Table D-5. OEM- Specific SEL Record

Byte	Data Field	Description
1-2	Record ID	
3	Record Type	
4-7	Time Stamp	
8-10	Manufacturer ID	

Table D-5. OEM- Specific SEL Record (continued)

Byte	Data Field	Description
11	Event Type	01h Login Event for Web 02h Logout Event for Web 03h Auto Logout Event for Web 04h Connection Lost Event for Web 09h Login Event for CLI/SSH/Telnet/LAN/LAN+ 0Ah Logout Event for CLI/SSH/Telnet/LAN/LAN+ 0Bh Auto Logout Event for CLI/SSH/Telnet/LAN/LAN+ 0Ch Connection Lost Event for CLI/SSH/Telnet/LAN/LAN+
12	User ID	For LDAP Maximum number of IPMI users + 1 For AD Maximum number of IPMI users + 2
13-16	IPv4 Address	All bytes will be 0xFF if IPv6 address is used.

D.4.2 Get Login Audit Configuration

This command returns the current login Audit configurations like enable/disable status of System Event Logs for RMCP/Web user login/logouts.

NetFn¹ 0x32
 Command 0x98

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Completion Code	Description
-	-	

Response Data

Byte	Completion Code	Description
1	00h	Success
2	Web Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable

Byte	Completion Code	Description
3	RMCP/RMCP+ Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable
4	Telnet Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable
5	SSH Audit Configuration	Bit 7:4 Reserved Bit 3 Enable/Disable Connection Lost Event 0 Disable 1 Enable Bit 2 Enable/Disable Auto Logout Event 0 Disable 1 Enable Bit 1 Enable/Disable Logout Event 0 Disable 1 Enable Bit 0 Enable/Disable Login Event 0 Disable 1 Enable

D.5 Get/Set SEL Policy Commands

D.5.1 Get SEL Policy Command

This command is used to get the current SEL policy (Linear/Circular).

NetFn ¹	0x32
Command	0x7E

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Not applicable

Response Data

Byte	Completion Code	Description
1	00h	Success
2	SEL Policy	00h Linear SEL 01h Circular SEL

D.5.2 Set SEL Policy Command

This command is used to set the SEL Policy.

NetFn ¹	0x32
Command	0x7F

1. This is the OEM function. Note that standard AMI OEM commands have adopted the 0x32 designation, while Tiler's set of command functions use 0x38.

Request Data

Byte	Data Field	Description
1	SEL Policy	00h Linear SEL 01h Circular SEL

Response Data

Byte	Completion Code	Description
1		00h Success CCh Invalid Data Field. D5h Not Support in Current State.

Appendix E: Supported Tileria OEM Commands

This section describes Tileria's commands, listed in [Table E-1](#), for the Baseboard Management Controller's (BMC's) Graphical User Interface (GUI) and Command Line Interface (CLI).

Note: In this implementation, there is no support for IPv6-related commands.

Table E-1. Tileria OEM Commands

Command Type	Commands
Module State Commands	• Get Module Status Command • Set Module State Command
Set Console Security Commands	• Set Console Security
Duty Cycle Commands	• Adjust Fan Settings Command
Switch Commands	• Switch USB Access Command • Switch I2C Access Command

E.1 Module State Commands

E.1.1 Get Module Status Command

This command queries the power and reset status of the CPU and I/O Modules.

NetFn ¹	0x38
Command	0x0

1. Tlera's set of command functions use the 0x38 NetFn designation, while the standard AMI OEM commands have adopted the 0x32 designation.

Example

```
raw 0x38 0x0
```

Request Data

Not applicable

Response Data

Byte	Function	Description
1		• 00h Success • 83h No Acknowledgment from Slave Device • FFh Unspecified Error
2	Query Power and Reset Status of CPU and IO Modules	Module Power Status
3	Query Power and Reset Status of CPU and IO Modules	Module Reset Status

E.1.2 Set Module State Command

NetFn¹ 0x38

Command 0x1

1. Tlera's set of command functions use the 0x38 NetFn designation, while the standard AMI OEM commands have adopted the 0x32 designation.

Request Data

Byte 1	Sub-Function	Description
0x0	00h Set Module Power	Module Mask for Commands 0 and 1: [7:4] IO Module Mask 0b is off/reset, 1b is on/running [3:0] CPU Module Mask 0b is off/reset, 1b is on/running
0x1	01h Set Module Reset	
0x2	02h Cycle Module Power	Module Mask for Commands 2 and 3: [7:4] IO Module Mask 0b is no action, 1b is cycle power/reset [3:0] CPU Module Mask 0b is no action, 1b is cycle power/reset
0x3	03h Cycle Module Reset	

Response Data

Byte	Function	Description
1		• 00h Success • 83h No Acknowledgment from Slave Device • C7h Invalid Length • C9h Parameter out of Range • FFh Unspecified Error

E.2 Set Power Cycle Interval Commands

E.2.1 Set Power Cycle Interval

NetFn¹ 0x38

Command 0x2

1. Tlera's set of command functions use the 0x38 NetFn designation, while the standard AMI OEM commands have adopted the 0x32 designation.

Request Data

Byte 1	Description
1	Interval in seconds

Response Data

Byte	Function	Description
1		• 00h Success • C7h Invalid Length

Example

Set the interval to 10 seconds.

```
-bash-3.2$ ipmitool -U admin -P admin -H 10.4.0.150 raw 0x38 2 10
```

E.3 Set Console Security Commands

E.3.1 Set Console Security

These commands enable BMC managers to specify if users need to use a password to log on to console accounts (for TILE-Gx™ Console access through the BMC, con1a etc...). If passwords are not required, access to the BMC is not restricted via passwords.

Currently, this is how to switch between the password required and password not required modes:

NetFunction	Command	Purpose	Operation	Op. Description
0x38	0x3	Set Console Security	0x1	No Password
0x38	0x3	Set Console Security	0x2	SSH Key Required ¹

1. Ensure that an SSH Key is uploaded using the GUI before running this command.

By default, no password is required to access the special TILE-Gx Console accounts. By starting an SSH or `tile-console` session to these accounts, a Kermit shell is started on the BMC to access the specified TILE-Gx Console.

NetFn ¹	0x38
Command	0x3

1. Tlera's set of command functions use the 0x38 NetFn designation, while the standard AMI OEM commands have adopted the 0x32 designation.

Note: Upload the SSH key for the `console` account before executing this command.

Request Data

Byte 1	Sub-Function	Description
0x1	00h	No Password Set the security level so passwords are not required in order to log in.
0x2	01h	SSH Key Required Set security level so only SSH keys are required and passwords are not required for logging in to BMC. Note: Upload the SSH Key for the <code>console</code> account before executing this command. Use The BMC to update the SSH as described in Uploading SSH keys on page 72 .

Response Data

Byte 1	Completion Code	Description
0x1	00h C7h C9h	Success Invalid Length Parameter out of Range

Example

Require an SSH-Key for TILE-Gx Console access.

```
-bash-3.2$ ipmitool -U admin -P admin -H 10.4.0.150 raw 0x38 3 1
```

E.3.2 Uploading Keys to the BMC via the GUI

1. If you require an SSH Key, you need to generate your own key. Refer to <http://en.wikipedia.org/wiki/Ssh-keygen> for instructions.
2. Login to the BMC via the browser. Use the pathname provided for you. We recommend that you use a current release of your browser.
3. At the logon screen (Figure E-1), enter the assigned username and password, and click **Login** to display the **Dashboard**.

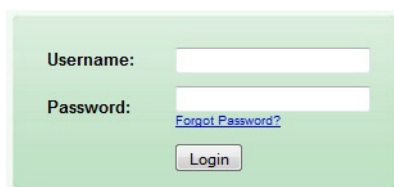


Figure E-1: Logon Screen

Note: To refresh the web display, should click the **Refresh** icon at the topmost-right menu bar (the third icon from right). Otherwise, clicking the browser's refresh button forces a logout.

- 4.
5. Upload the key to the BMC via the GUI by:
 - a. Selecting **Configuration→Users**, and the **console** user
 - b. Clicking on **Modify User**.
 - c. Selecting a **New SSH Key** by browsing for the file.

You can also add this key to any User Account you use or create.

Note: The only method available for uploading SSH keys is via the Web GUI.

Modify User

Username: admin

☐ Change Password

Password Size: ☐ 16 Bytes ☐ 20 Bytes

Password:

Confirm Password:

User Access: ☒ Enable

Network Privilege: Administrator

SNMP Status: ☐ Enable

SNMP Access: Read Only

Authentication Protocol: SHA

Privacy Protocol: DES

Email ID:

Email Format: AMI-Format

Uploaded SSH Key: Not Available

New SSH Key: No file selected.

Figure 6. Modify User Screen

E.4 Duty Cycle Commands

E.4.1 Adjust Fan Settings Command

This command adjusts the initial duty cycle value used for the automatic fan control algorithm, allowing limited control over fan speed.

Note: This command does not persist across BMC restart or system reboot.

NetFn¹ 0x38

Command 0x80

1. Tlera's set of command functions use the 0x38 NetFn designation, while the standard AMI OEM commands have adopted the 0x32 designation.

Request Data

Byte	Description
1	Percentage Duty Cycle 0 - 100

Response Data

Byte	Sub-Function	Description
		• 00h Success • C7h Invalid Length • C9h Parameter out of Range

Example

For Set Starting Duty Cycle Command, cycle all CPUs' power:

```
-bash-3.2$ ipmitool -U admin -P admin -H 10.4.0.150 raw 0x38 0x80 15
```

E.5 Switch Commands

E.5.1 Switch USB Access Command

Note: This command applies only to the TILEmpower-Gx36 FR platform.

NetFn ¹	0x38
Command	0xF5

1. Tlera's set of command functions use the 0x38 NetFn designation, while the standard AMI OEM commands have adopted the 0x32 designation.

Request Data

Byte 1	Data Field
0	• 00h Disable USB • 01h USB access via the BMC • 02h USB access via the internal USB debug port

Response Data

Byte 1	Data Field
0	• 00h Success • 83h No Acknowledgment from Slave Device • C7h Invalid Length

Example

Enable USB access via BMC usb proxy (see `tile-monitor` usage for more information).

```
-bash-3.2$ ipmitool -U admin -P admin -H 10.4.0.150 raw 0x38 0xf5 1
```

E.5.2 Switch I²C Access Command

.

NetFn ¹	0x38
Command	0xF0

1. Tlera's set of command functions use the 0x38 NetFn designation, while the standard AMI OEM commands have adopted the 0x32 designation.

Request Data

Byte	Data Field
1	TILE-Gx masters shared I ² C bus.
0	BMC masters shared I ² C bus.

Response Data

Not applicable

APPENDIX F: GLOSSARY

Term	Definition
BMC	Baseboard Management Controller. The BMC is a specialized micro-controller installed on the motherboard of a computer, which manages the interface between system management software and the hardware. In its role, the BMC monitors sensors on the platform and reports activity, including temperature and general operating states.
I/O	Input/Output.
ipmitool	The utility for controlling IPMI-enabled devices.
IPMI	Intelligent Platform Management Interface.
IP network	Internet Protocol network. The principal communications protocol in the Internet protocol suite.
iptables	A user space application program that allows a system administrator to configure the tables provided by the Linux kernel firewall (implemented as different Netfilter modules) and the chains and rules it stores. Different kernel modules and programs are currently used for different protocols; iptables applies to IPv4.
LDAP	Lightweight Directory Access Protocol. An open, vendor-neutral, industry standard application protocol for accessing and maintaining distributed directory information services over an IP network.
MDE™	Multicore Development Environment™. Tiler's multicore programming environment.
Netfilter	A set of hooks inside the Linux kernel that allows kernel modules to register call-back functions with the network stack.
PEF	Platform Event Filtering. Provides a mechanism for configuring the BMC to take selected actions on event messages that it receives or has internally generated. These actions include operations such as system power-off, system reset, as well as triggering the generation of an alert.
ssh-keygen	A Unix utility used to generate, manage, and convert authentication keys for ssh authentication.
SSH	Secure Shell. A network protocol using cryptographic methods to ensure secure data communication, remote (CLI) logins, remote command execution, and other secure network services.
SSL	The Secure Socket Layer protocol was created by Netscape to ensure secure transactions between Web servers and browsers. The protocol uses a third party, a Certificate Authority (CA), to identify one end or both end of the transactions.
YAFUFlash	Yet Another Firmware Upgrade Flash. A BMC firmware update tool that allows users to flash the BMC via network or USB connections. Yafuflash is available for both Linux and Windows-based operating systems.

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